

CL-900i(EJK024) Series CLIA

Installation Guidance



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Chapter 1 Check before Installation

1.1 Installation Environment

- Altitude: -400-3,000m.
- The system is for indoor use only.
- The bearing platform (or ground) should be level (with gradient less than 1/200).
- The bearing platform (or ground) should be able to support at least 160Kg weight.
- The installation site should be well ventilated.
- The installation site should be free of dust.
- The installation side should not be in direct sun.
- The installation site should be kept away from a heat or draft source.
- The installation site should be free of corrosive gas and flammable gas.
- The bearing platform (or ground) should be free of vibration.
- Operating temperature: 15°C-30°C with fluctuation <2°C/H. Provide air conditioning equipment if the room temperature does not meet the requirements.
- Relative humidity: 35%-85% RH, without condensation.

1.2 Space and Accessibility Requirements for Unpacking

- Dimensions of whole unit after packing: 1130 mm(length)×910mm(depth)×760mm(height)
- Unpacking space requirement: After putting down the instrument with the packing box, the distance from the wall is ≥ 0.5 m for unpacking operation.

1.2.1 Space and Accessibility Requirements for Installation

- Dimensions of whole unit before packing: 860mm(length)×740 mm(depth)×560mm(height).

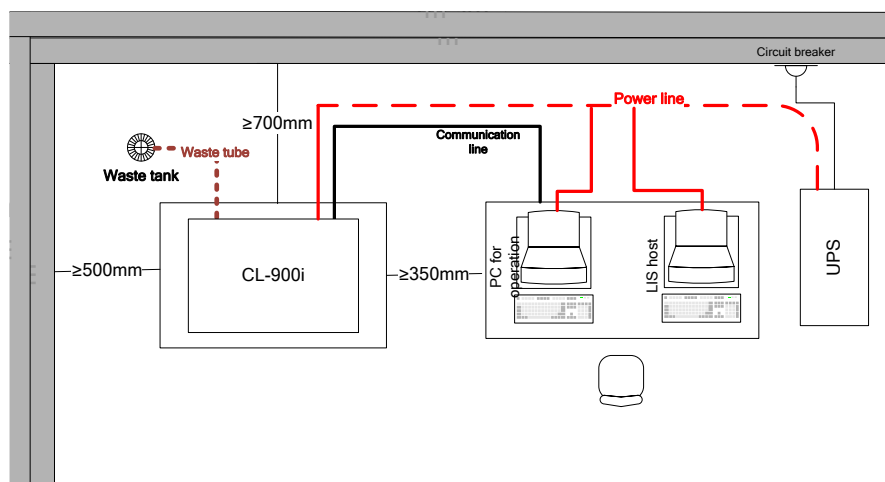


Figure 1-1 Space and accessibility requirements for installation of the CL-900i

NOTE: On the right of the instrument required not less 350mm space for maintain the filter.

1.2.2 Power Supply and Noise

- Power supply: 100V-240V~ 50Hz, 100V/240V~ 60Hz. Voltage fluctuation: +/-10%. Line frequency: +/-1Hz. Three-wire power cord with good grounding performance.



WARNING

Make sure the power socket is grounded correctly. Incorrect grounding may lead to electric shock or equipment damage.

Check if the power socket outputs voltage meeting the specified requirements and has a proper fuse installed.

-
- Rated input power of analyzer: 1000VA. The instrument should be connected to a power socket with load no less than 2.5A.
 - The ground voltage should be <5V, be sure the ground voltage is only for the instrument.

How to make sure the ground voltage:

The voltage of between neutral wire and live wire similar the voltage of between neutral wire and ground wire, also less than the standard voltage.

- If the user is going to use a UPS to power the instrument, make sure that the UPS can provide power supply greater than or equal to 2000VA (analyzer + computer + printer).
- The installation site should be kept away from big noise and power supply interference.
- Keep the system away from brush-type motors and electrical contact devices that are frequently switched on and off.
- Do not use such devices as mobile phones and radio transmitters near the system.

1.2.3 Drainage Check (if draining water through a sewer)

- The chemiluminescence immunoassay analyzer can directly discharge the waste liquid to the sewer of special waste liquid treatment. The length of the waste tube is less than 5 m and greater than 1 m.
- The waste outlet should be no less than 16mm wide.
- The waste outlet should be no higher than 50cm.



BIOHAZARD

Dispose of liquid waste according to the local regulations.

1.2.4 Recommended PC Configuration

Item	Description
CPU	At least Intel P4, 2.8GHz
Random access memory (RAM)	At least 4GB

Item	Description
Network adapter	The computer is connected to the analyzer through a network adapter. If you are going to connect the computer with the LIS or Internet, you should prepare another network adapter (Intel gigabit network adapter)
Hard disk defragment	At least 500GB, with SATA interface. Install the operating system in the C drive and the operating software of the instrument in the D drive. Make sure that the C drive is over 30G and D drive over 100G, and the disk file system is of NTFS format. Deselect the two options at the bottom of the disk properties window: "Compress drive to save disk space" and "Allow Indexing Service to index this disk for fast file searching" .
Operating system	The operating system installed on the computer must be an activated Microsoft Windows 10 HOME 1903 or a previous version of the operating system
Application software	Except for the operating system, other application software must not be installed or reserved on the computer. If an anti-virus application has been installed, then remove the automatic scheduled scanning and add the operating software and BSLOG to the trust list.
Screen saver and system standby	Turn off the screen saver and BS Special Power Policy power scheme, and then disable the hibernation option.
Screen display properties	17" touchscreen monitor or above, 16:9 or 4:3 with resolution of 1280×1024.
Automatic synchronization with Internet time server	Disable the Automatically synchronize with an Internet time server option.
Automatic updates	Turn off the automatic updates.
Auto startup setup	If you are going to use the auto startup function, perform necessary settings for BIOS and network adapters while referring to their operation manuals.

1.3 Configuration Check

Table 1 Configuration list

• CL-900i/CL-960i Chemiluminescence Immunoassay Analyzer -----	1
• Accessory kit-----	1
• Waste tank -----	2
• Optional modules	
Printer -----	1

Chapter 2 Instrument installation

2.1 Tools

Hexagon wrenches (M3~M6), 13# box spanner, tape measure, 200mm spanner, cross screwdriver, flathead screwdriver and snap-off knife



Figure2-1Tools

2.2 Installation Procedure

2.2.1 Unpacking

- 1) Check against packing list: Check the instrument configuration against the packing list before unpacking according to section 1.3 System Configuration.



Figure2-2Serial No. and packing list

- 2) **Unloading:** Unload the instrument from the truck. It is recommended to use a forklift.

Note: There is a tilt warning label on the main unit package, which can change in color for tilting. When the center of the label turns red, it means the package has been seriously tilted and instrument damage may be caused. Contact the logistics company immediately to prevent unexpected results.



Figure2-3Tilt warning label in normal conditions

- 3) **Transfer:** Use a forklift to transport the instrument to a location suitable for unpacking.

Note: Keep the proper distance between the instrument and the wall to facilitate the removal of the package.



Figure2-4Transferring Instrument to a Location Suitable for Unpacking

- 4) **Unpacking analyzer:** Cut the packing ropes with scissors, remove the upper cover of the main box, →remove the protective plates on both sides, →and remove the boarding. Deliver the removed packaging materials to the customer for safekeeping.

**NOTE**

After unpacking, remind the user to keep the packaging materials for 3 months for use in returning or replacing the instrument.



Figure2-5Outer Package Removed

- 5) Remove the instrument protective film and accessory foam.

**CAUTION**

After remove the accessory foam, remind the user to keep the accessory foam for 3 months for use in returning or replacing the instrument.



Figure2-6Accessory foam removed

6) Carry the analyzer to the desk:

- a) Confirm that the carrying handles at the four ends of the instrument have been tightened;
- b) Lift the instrument and move it to the test bench by four persons;

**CAUTION**

Because the analyzer is heavy, it is recommended to carry it 4 people together and walk with caution.

Be careful of bruises! Be careful not to hit other fragile items! Be careful not to scratch the instrument!



Figure2-7Carry the analyzer to the desk

- c) After the instrument is placed on the test bench, adjust the anchor of the instrument to make the machine evenly bear the force;

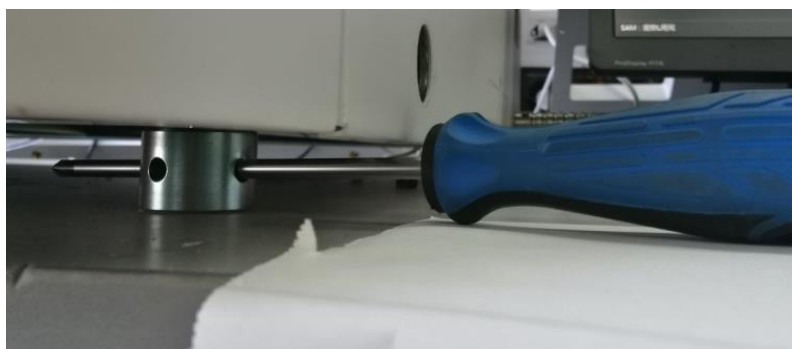


Figure2-8Adjusting the anchor of the Instrument

Note:

- CI-900i has only one adjustable foot, on the left side of the front end of the instrument. Therefore, the client needs to confirm the flatness of the desktop to ensure that the installation of the instrument meets the requirements of inclination, otherwise it is difficult to adjust the level of the instrument.
- In the process of installation, if the instrument is found unstable and needs to be adjusted to raise the adjustable footing, please remove the adjustable footing and remove the above two flat gaskets, and then adjust the height after installing the footing back, as shown in the figure below.

**7) Removing Clear protective accessories:**

- Unscrew the carrying handles of the instrument and hand them to the customer for safekeeping for subsequent transfer;
- Remove all the fixing tapes on the surface of the instrument;
- Open the front door, and remove the protective foam of the front door;



Figure2-9 Remove the protective foam of the front door

- Remove the screws of the transparent cover to open it;

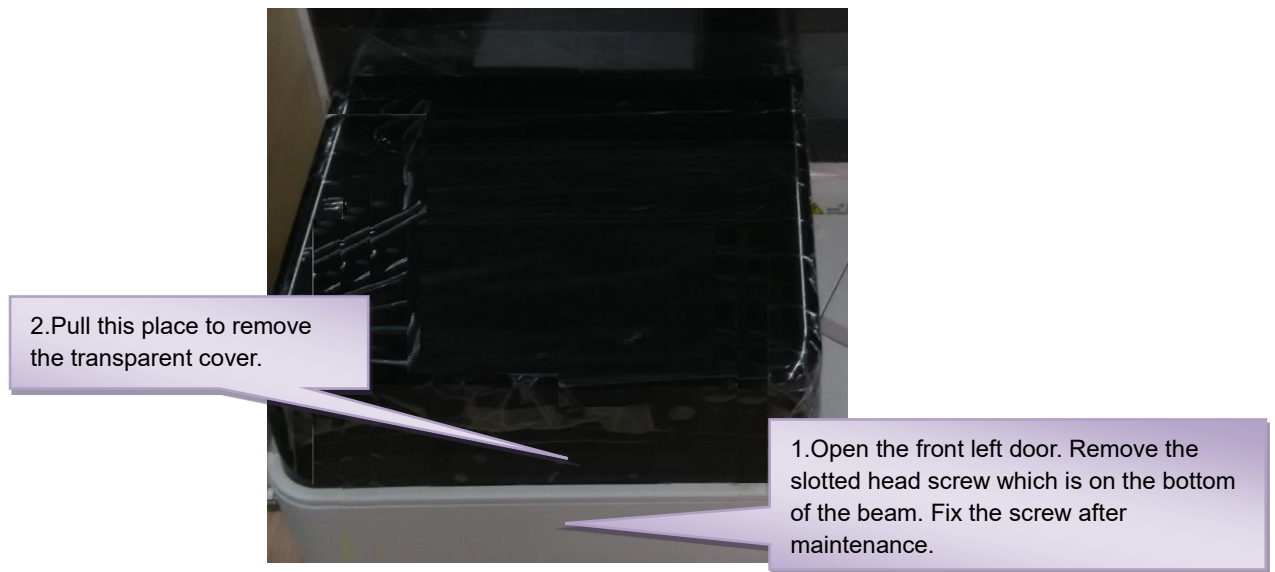


Figure2-10 Remove the screws of the transparent cover



Figure2-11 Remove the transparent cover

- e) Remove all fixing plates as follow :
- f) Remove the fixing plate for fixing gripper Y axis. First, remove the screw from the position below;

Remove the fixing plate.

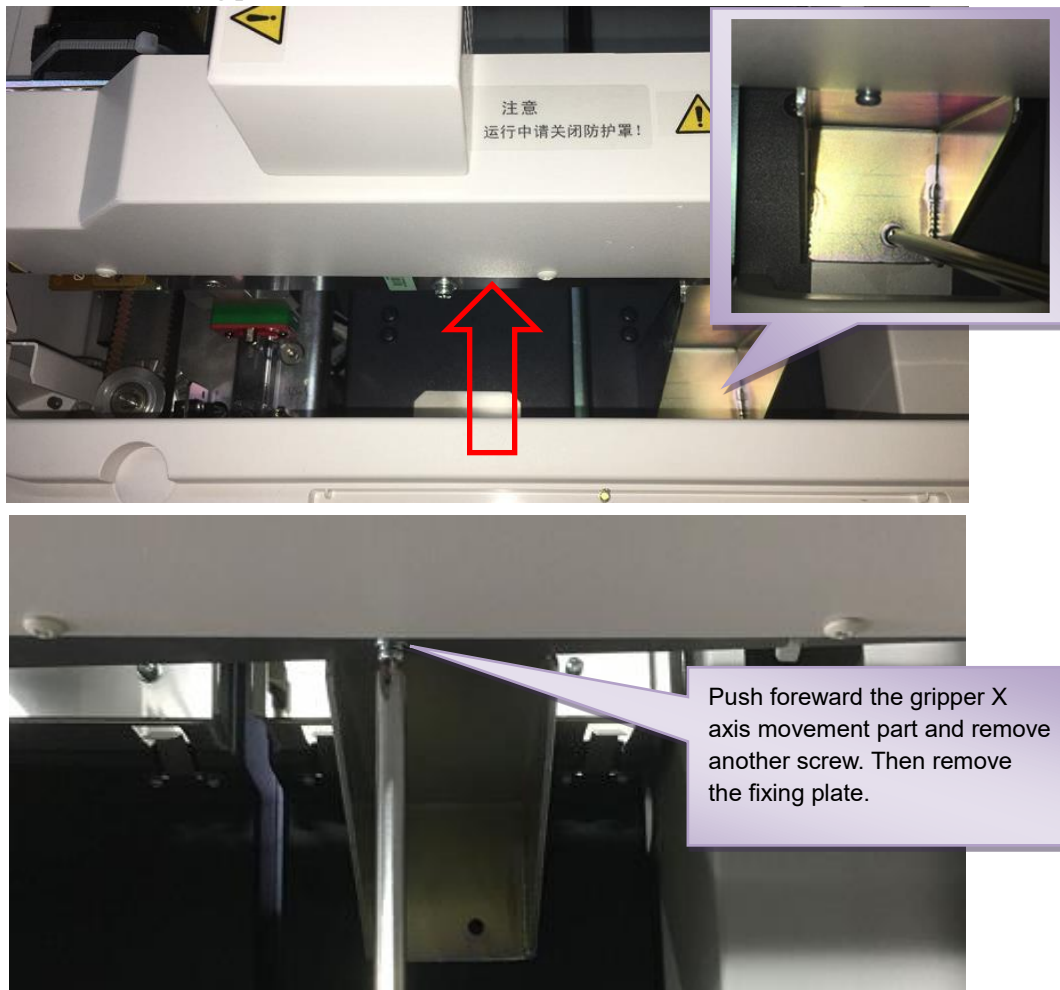


Figure2-12 Remove the screw from the position below

g) Remove the fixing plate for the drawer assembly

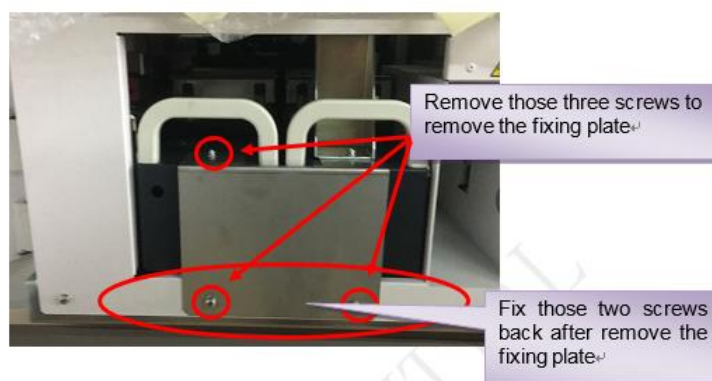


Figure2-13 Remove the fixing plate for the drawer assembly

h) Remove the fixing plate for fixing gripper X axis

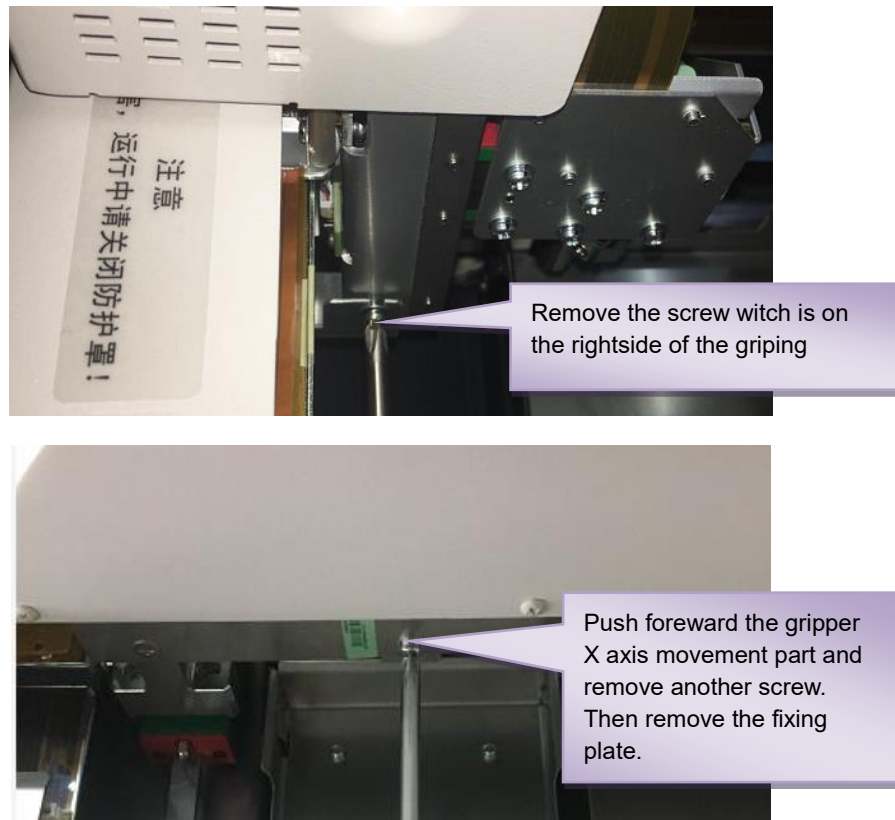


Figure2-14 Remove the fixing plate for fixing gripper X axis

- i) Remove the fixing plate for fixing gripper Z axis

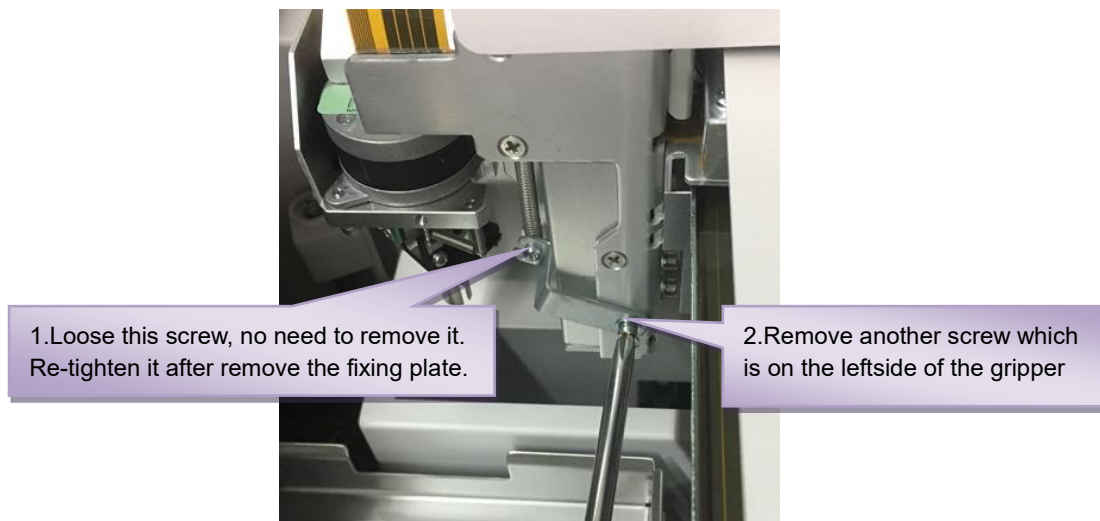


Figure2-15 Remove the fixing plate for fixing gripper Z axis

- j) Unplug the rubber covers on both sides, remove the hex socket screws on both sides, and press the top two sides of the front vertical plate. When removing the pop-up, take care not to pull the wiring on both sides.



Figure2-16The rubber covers

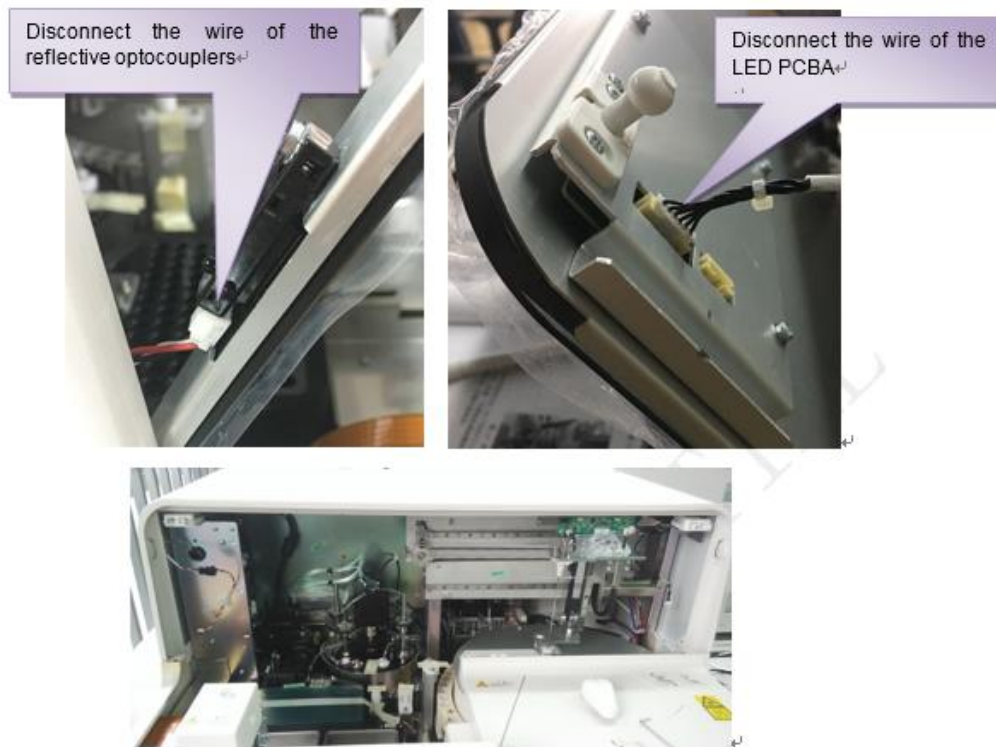


Figure2-17Removing the Front Vertical Plate

- k) Remove the fixing plate of the Probe drive assembly

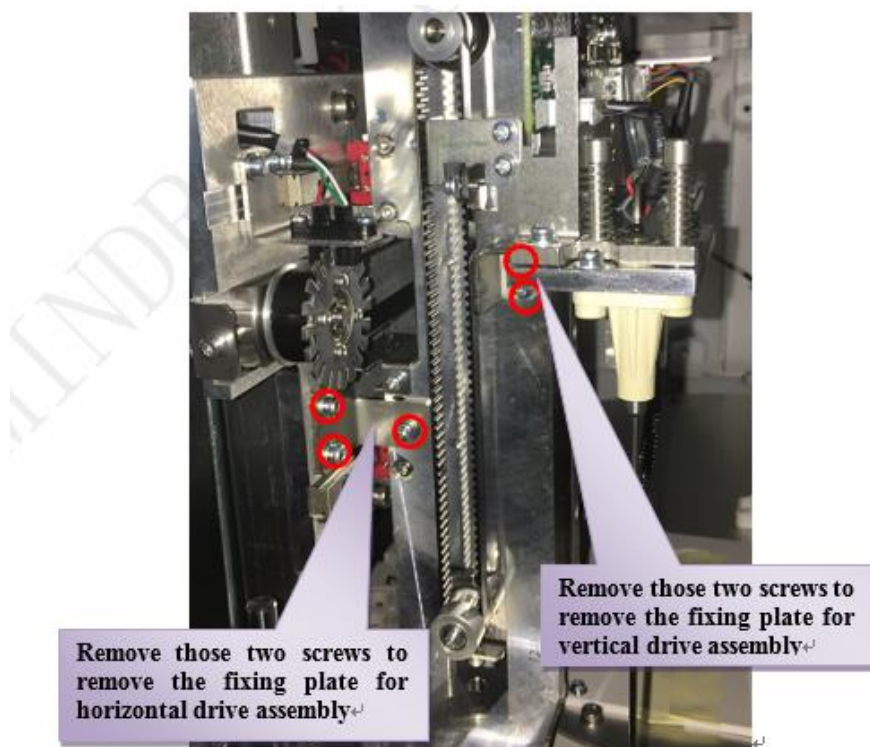


Figure2-18 Remove the fixing plate for the probe drive assembly

- l) Remove the fixing plate for waste drainage

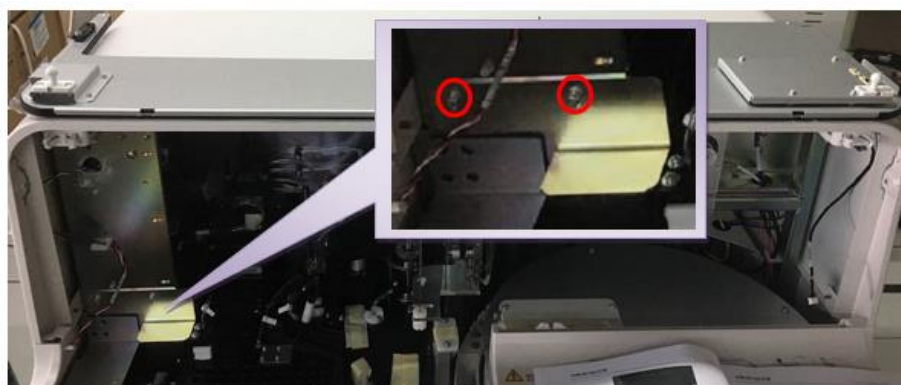


Figure2-19 Remove the fixing plate for waste drainage

- m) Remove the sealing tapes around the incubation block, dispersion, mixing position, wash well, and aspirate plate of the reagent chamber in turn, and keep the small hole opaque disk of the incubation block for subsequent use.

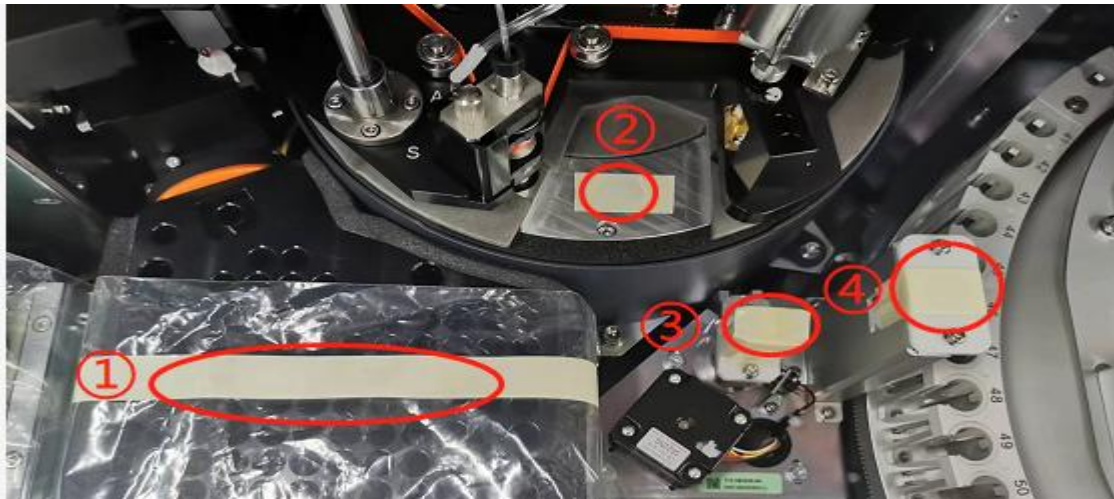


Figure2-20Sealing Tapes around the Incubation Block, Dispersion, Mixing Position, and Wash Well

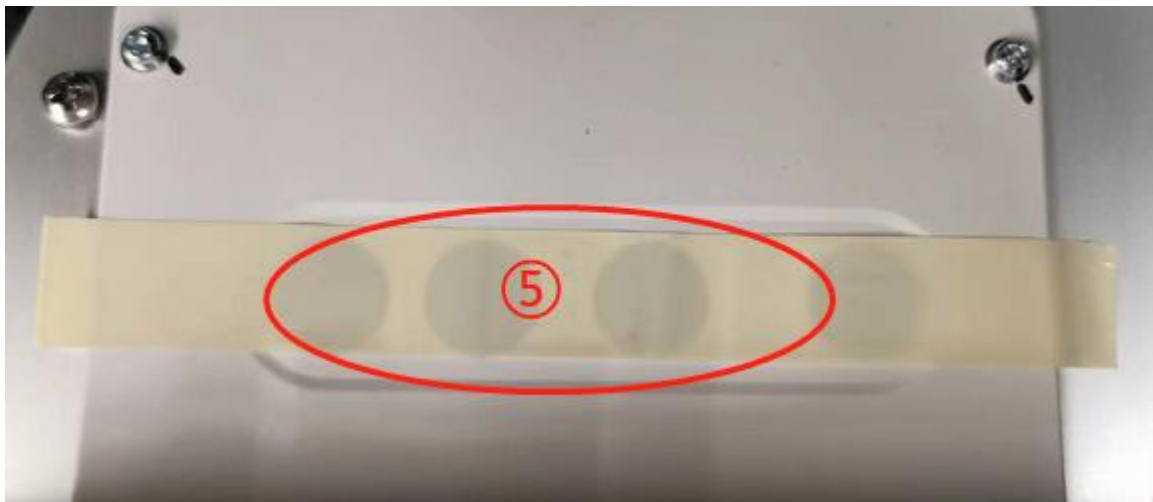


Figure2-21Sealing Tapes around the Aspirate Plate of the Reagent Chamber

n) There are 7 fixing plates, make sure all fixing plates are removed

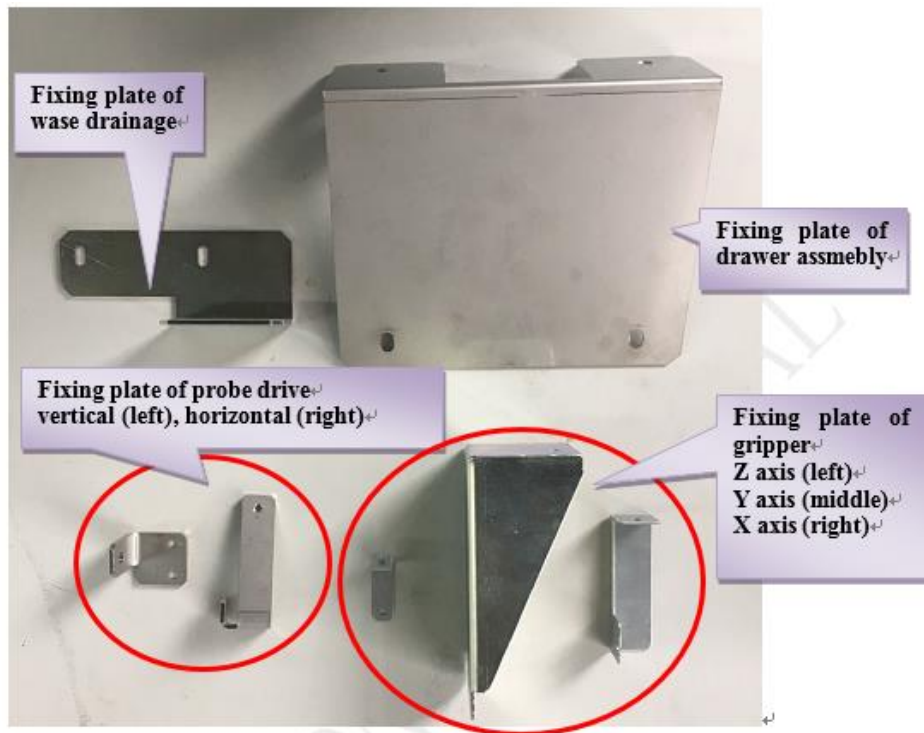


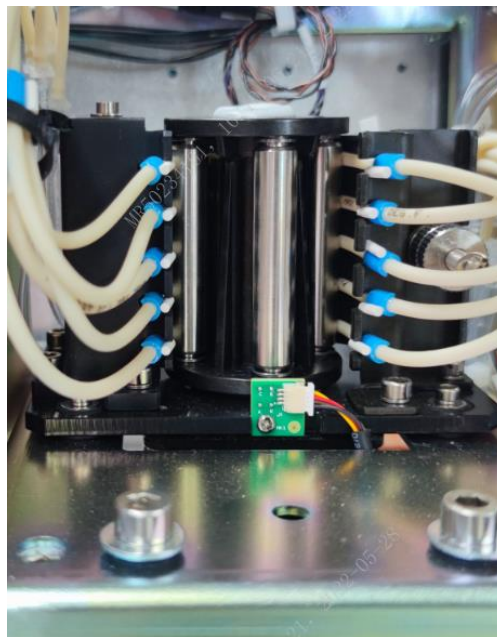
Figure2-22All the fixing plates

Note: It is recommended to give the above fixed parts and handle to the customer for temporary storage, so as to facilitate transfer and other business use for future.

2.2.2 Fluidic Connection

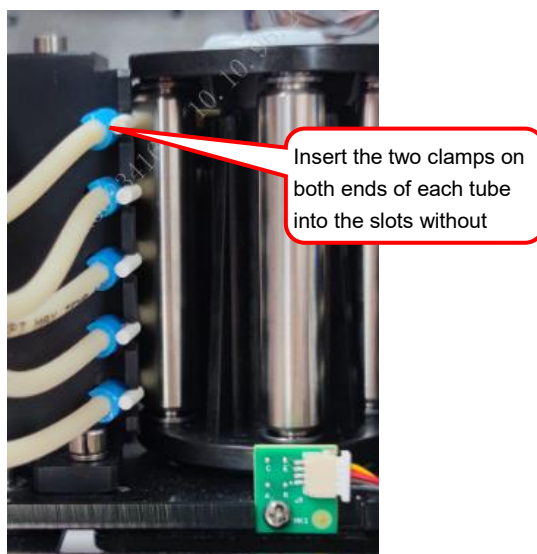
2.2.2.1 Install Peristaltic Pump Tubes

- 1) Remove the pump maintenance window cover to find the dispersion peristaltic pump assembly.

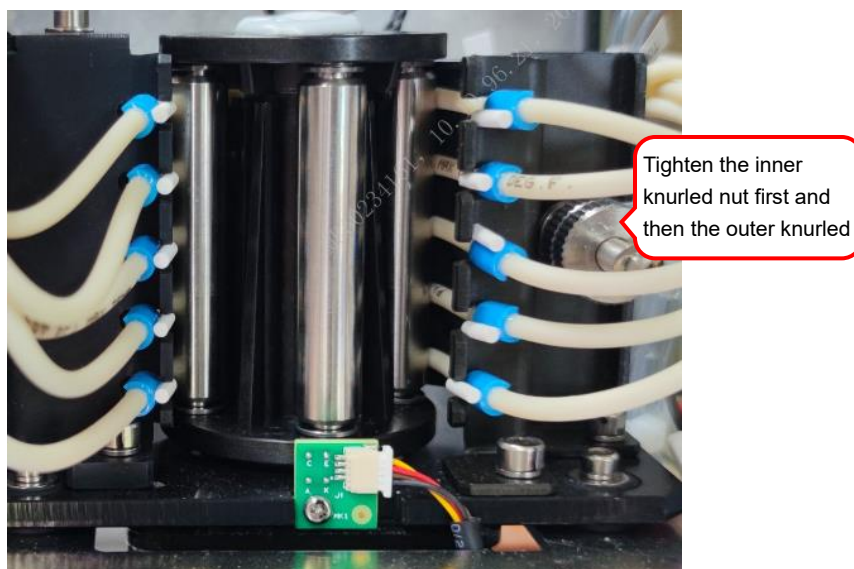


- 2) Insert the two clamps on both ends of the 5 peristaltic pump tubes into the slots, and make sure that the clamps are tightened without deviation. Install the tubes from bottom to top. After installing each pump tube, hold the upper flange of the roller and rotate the roller clockwise at least 10 times to ensure that

the pump tubes are in their respective channels to avoid intertwining.



- 3) After fixing the pump tubes, tighten the inner knurled nut first and then the outer knurled nut. Do not tighten the two nuts simultaneously.



- 4) Check if the pump tubes are installed correctly and if the pump tubes are twined, pulled, or interfered with the inlet and outlet tubes. If no problem is found, the dispersion peristaltic pump tubes installation is finished.

Note:

- The clamps at both ends of the five pump tubes are clamped tightly in the slots without deviation.
 - Tighten the inner nut close to the shaft shoulder and then tighten the outer nut and the first screw. Do not tighten the two nuts at the same time.
- Make sure the direction of all tubes are correct. Otherwise, dispersion wash may fail. If you cannot confirm the direction, refer to other unremoved tubes.

2.2.2.2 Connecting Waste Tubes

The instrument has two waste drainage outlets: waste outlet 1 and waste outlet 2. According to

the actual situation of the customer, you can choose to use the waste tank, or directly discharge the waste.

Use the waste tank

- 1) Take out the waste tank cover assembly and the waste sensor assembly, and connect one end of the two waste tubes to “waste outlet 1” and “waste outlet 2” of the instrument, and connect the waste sensor at the same time.

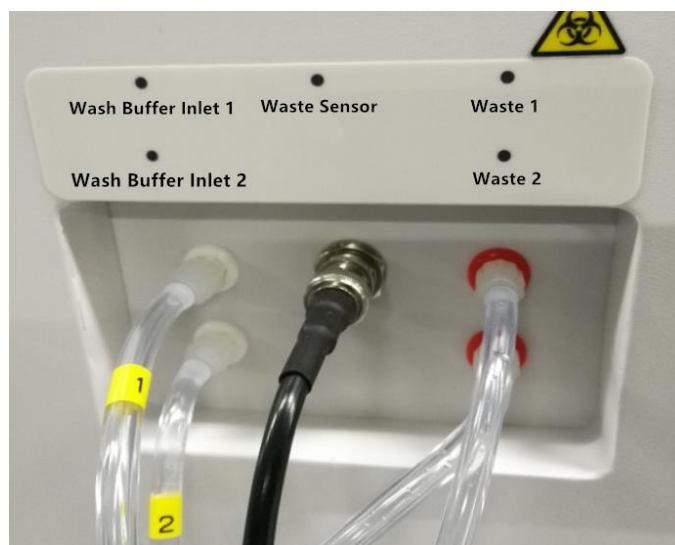


Figure2-23Waste Tube and Waste Sensor Connection

- 2) Connect the other end of the waste tubes to the waste tank cover and install the waste sensor.



Figure2-24Waste tank connection

- 3) Place the waste tank and complete the connection of the waste tank.

NOTE: Do not connect the tubes of waste outlet 1 and waste outlet 2, which will cause waste to be flowed backward!

Directly discharge the waste

Enter the interface of **Utility->System Setup** and select “**draining of liquid waste**”.

If the waste can be discharged through sewer, follow the steps below:

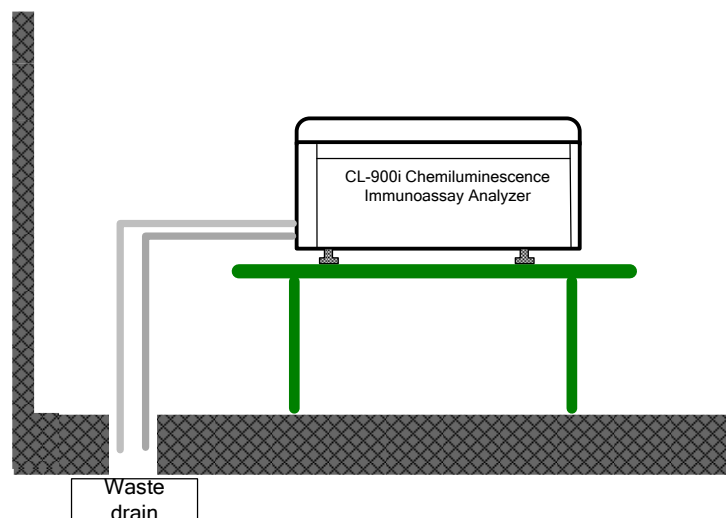


Figure2-25Waste discharge

NOTE: Do not connect the tubes of waste outlet 1 and waste outlet 2, which will cause waste to be flowed backward!

Requirements for connecting waste tubes:

- 1) Insert the waste tube into the sewer, and make sure that the tube is about 1m~5m long.
- 2) Make sure that the waste outlet is no less than 50cm high.
- 3) Cover the waste outlet to eliminate noise and prevent bubbles overflow.
- 4) When installing the tubes, prevent them from being bent, twisted, or pressed, and keep them away from any sharp-edged objects or others that may scratch the tubes.
- 5) Ensure that all electric connectors are away from and higher than hydropneumatic system parts.
- 6) Ensure that all electric connectors are higher than and away from fluidic parts and prevent fluidic connectors from facing electric components.



BIOHAZARD

Dispose of liquid waste according to the local regulations.

2.2.2.3 Connect Wash buffer tank

- 1) Take out the wash buffer tank assembly.
- 2) Connect the tubes of the wash buffer tank assembly to the wash buffer inlets of the instrument;

Note: The tube with mark 1 is connected to wash buffer tank 1; the tube with mark 2 is connected to wash buffer tank 2. Do not mistake them!

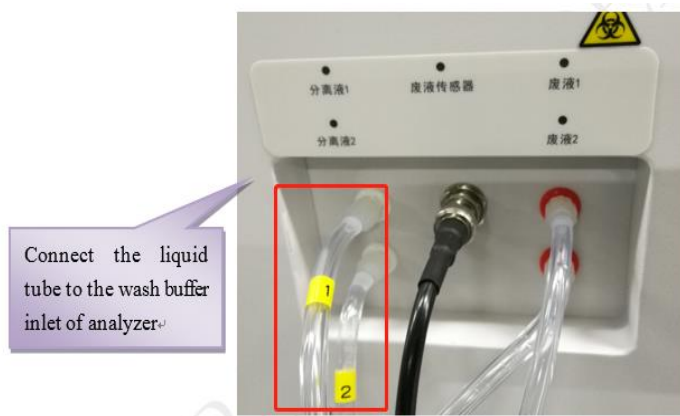


Figure2-26Wash Buffer Tube Connection

- 3) Take two wash buffer tanks. Remove the cover of the wash buffer tank and place the wash buffer tank assembly into the tank and tighten the cover.



Figure2-27Placing the Wash Buffer Tank Cover Assembly into the Tank and Tighten the Cover

- 4) Place the wash buffer tank and complete the connection of the wash buffer tank.

2.2.2.4 Final Connection Effects



Figure 2-28 Final connection effects

The layout of the waste tank and the wash buffer tank can be adjusted slightly according to the actual situation of the customer.

2.2.3 Connect the computer

Connect the PC with analyzer

The network cable is used to connect the analyzing unit. The IP of the network adapter on the PC should be set as **192.168.23.3**.

2.2.3.1 Analyzer Connection

External cables to be connected to the analyzer end:

- Power cord: used for powering the analyzer;
- Network cable: used for communication with the computer.

External cables to be connected to the computer end:

- Computer power cord;
- VGA cable;
- Hand-held barcode scanner;
- Network cable used for communication with the analyzer.

The LIS cable has two connection modes: Serial port and network port, which can be used according to site requirements.



Figure 2-29 Analyzer Connection

2.2.4 Software Installation

Complete the software installation according to the SetupGuide. The precautions for software installation are listed below:

2.2.4.1 Software version requirements:

Before installing the software, please confirm that the software installation package is the latest version!

2.2.4.2 Upgrade control software

- a) For the first installation or the first upgrade of CL900i, please to select the **Instrument type** in the UpgradeTool first, After selecting the Instrument Type, the upgrade details will be displayed, and then click to download,as shown below:

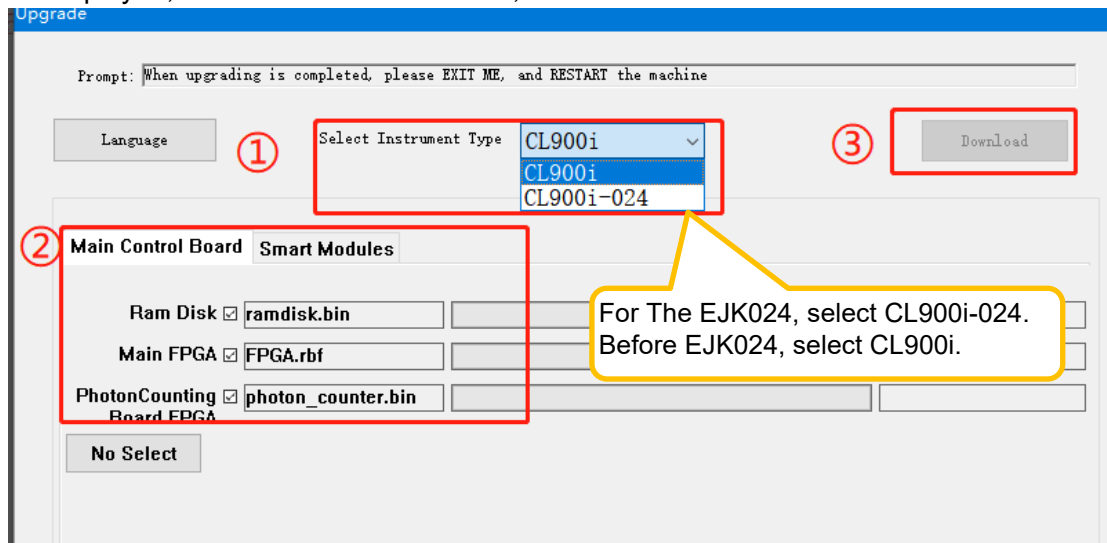


Figure 2-30 For the first installation or the first upgrade

- b) If it is not installed for the first time or upgraded again, the default analyzer type will be displayed, please confirm that it is consistent with the actual model, and then click Download. as shown below:

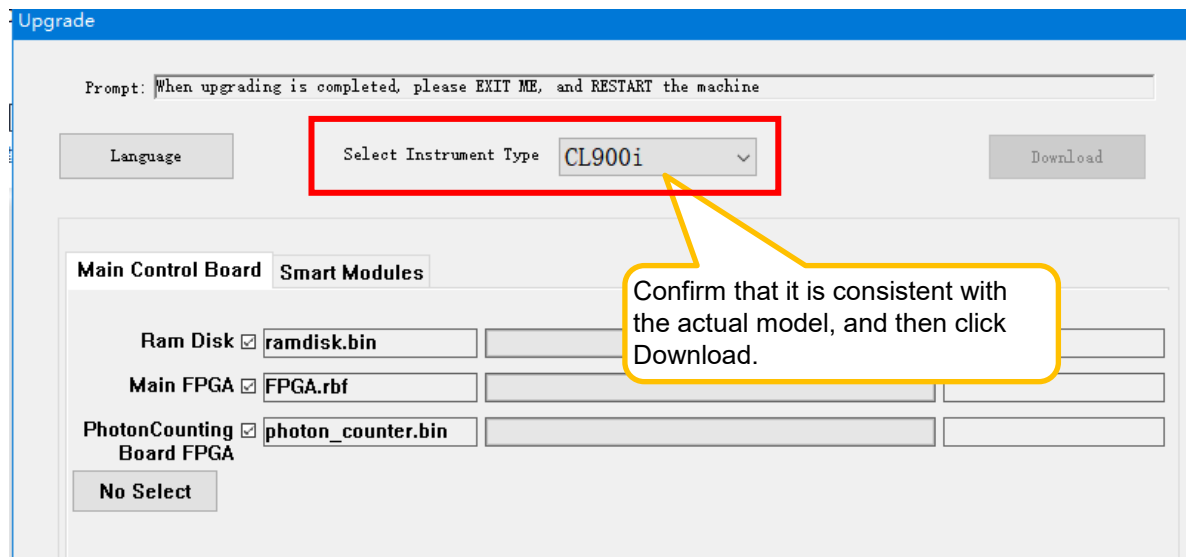


Figure 2-31 Not the first installation or upgrade again

Note:

- a) If the instrument type is not set as required, the following prompt will pop up when entering the operation software. Please use UpgradeTool to correctly configure the instrument type and then upgrade the control software:

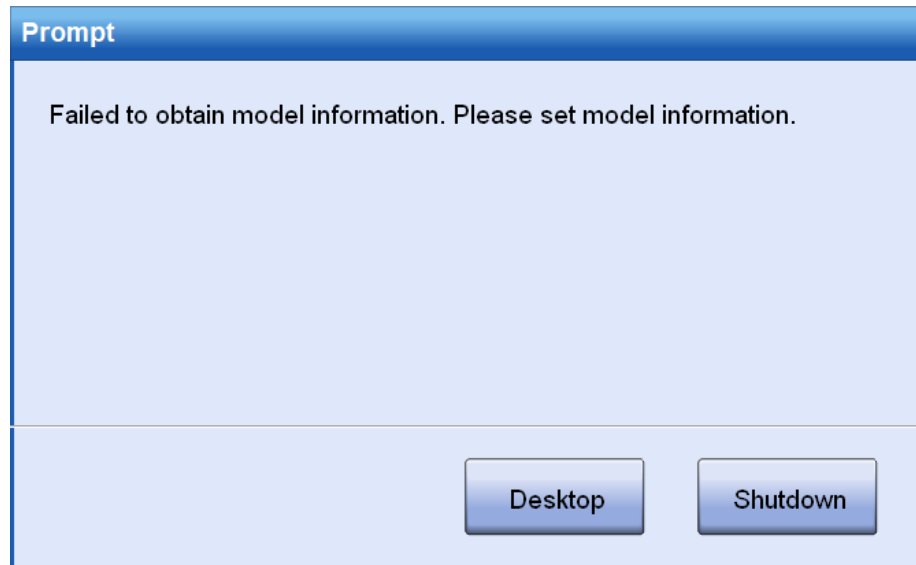


Figure 2-32 Failed to obtain model information

- b) If the selected type does not match the actual type when upgrading the control software, the following prompt will pop up when entering the operation software. Please use UpgradeTool to select the correct instrument type to upgrade again:

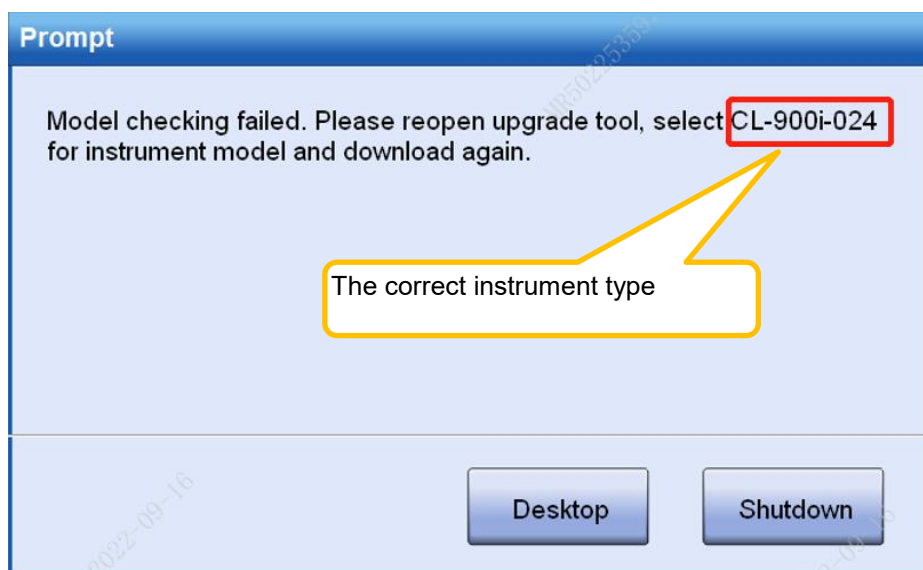


Figure 2-30 Model checking failed

Chapter 3 Power on and alignment

3.1 Preparation for Powering On

- 1) Load solid waste container.

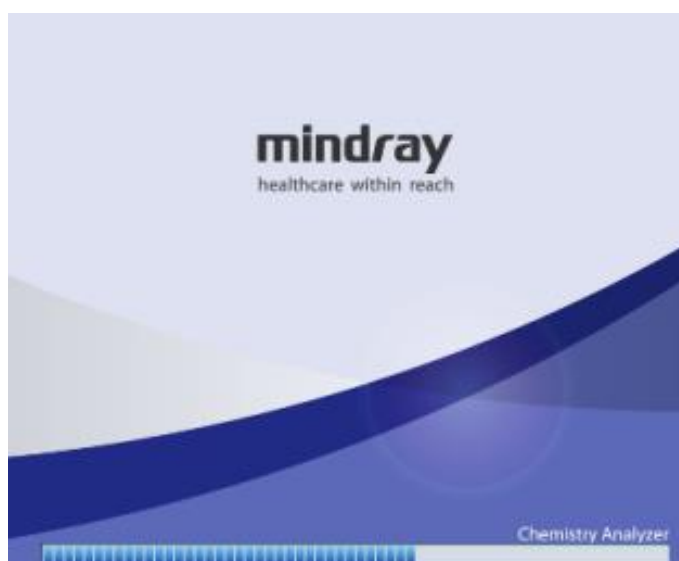


Figure3-1Load solid waste container

- 2) Be sure had installed the front vertical plate.
- 3) Power on the instrument

3.2 Initial Startup

1. Refer to the "Setup Guide" in the software installation package to complete the upgrade of the control software, and restart the computer and instrument after completing the upgrade.
2. Start up the computer. The CL-900i Operating Software is run automatically. See the figure below.



3. Enter the username "ServiceUser" and password "#BS8A#SEU" in the login window



1. Load the Probe Cleanser (DC) and Deionized water(DI Water)

- 1) Select **Reagent>Consumable Management**;

Figure3-4Consumable management screen

- 2) Select **Probe Cleanser (DC)** , tap **Load**, input the corresponding information and margin of Probe Cleanser, and load the Probe Cleanser

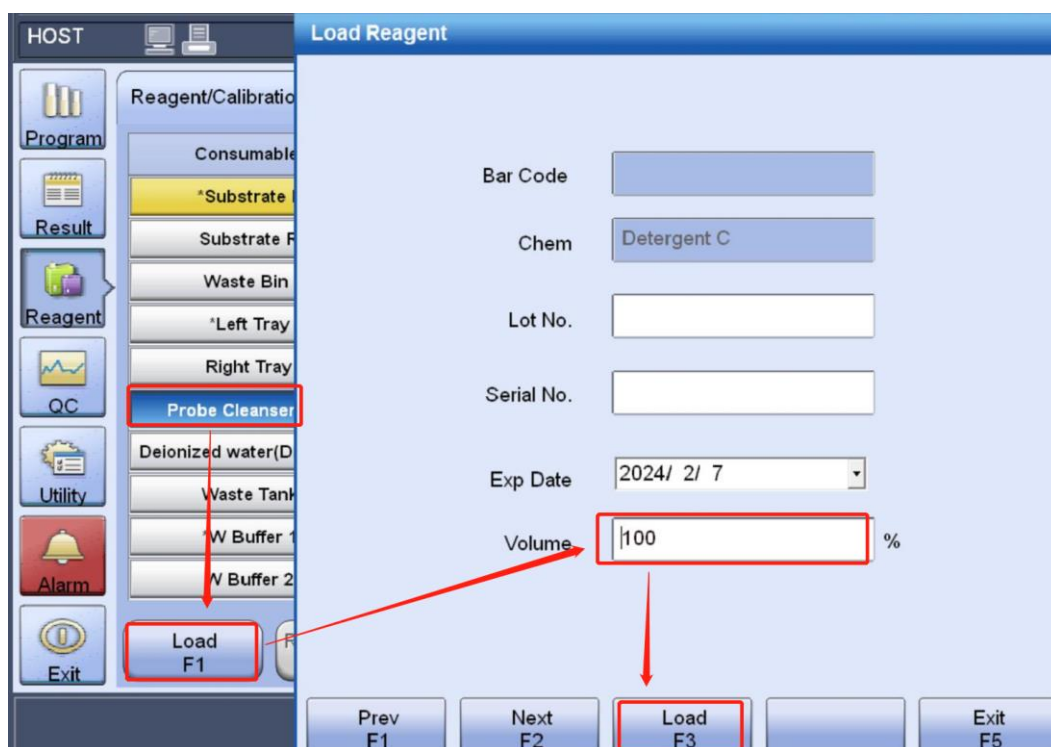


Figure3-5 Load the Probe Cleanser (DC)

- 3) In the same way, Select **Deionized water(DI Water)**, tap **Load**, input the margin of Deionized water, and load the Deionized water.

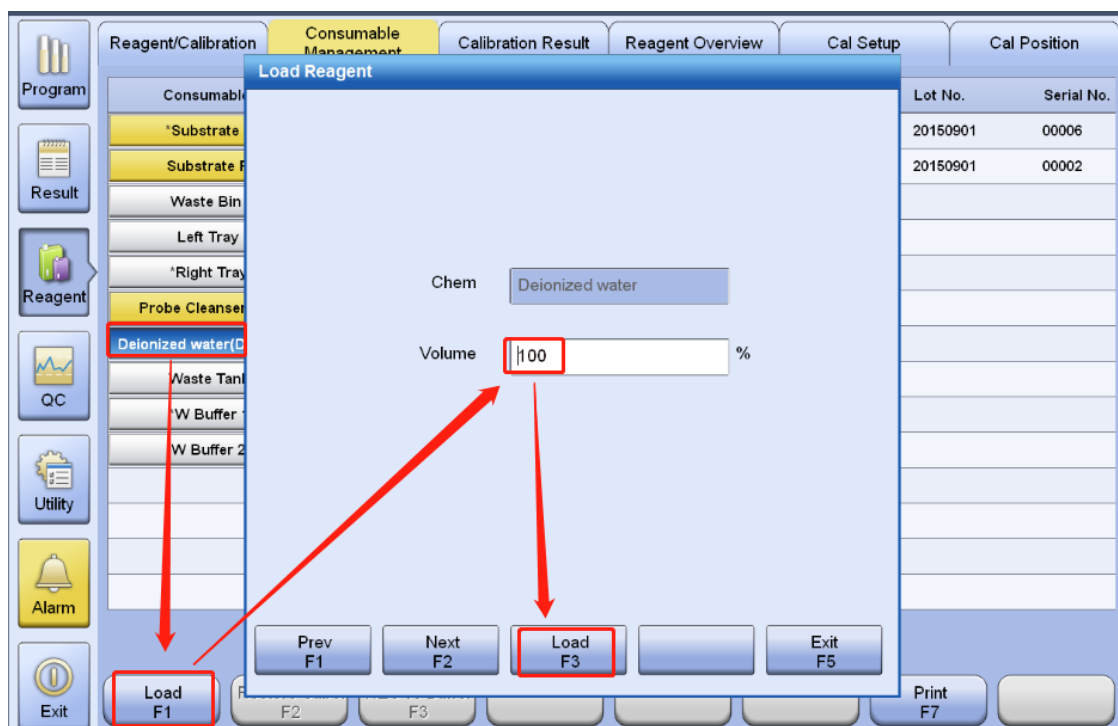


Figure3-6 Load the Deionized water

2. Load Wash Buffer and prime Tubes

- 1) In the **Consumables Management** screen, select **Wash Buffer 1** and **Wash Buffer 2**, respectively, and tap **Load** to load the wash buffer;

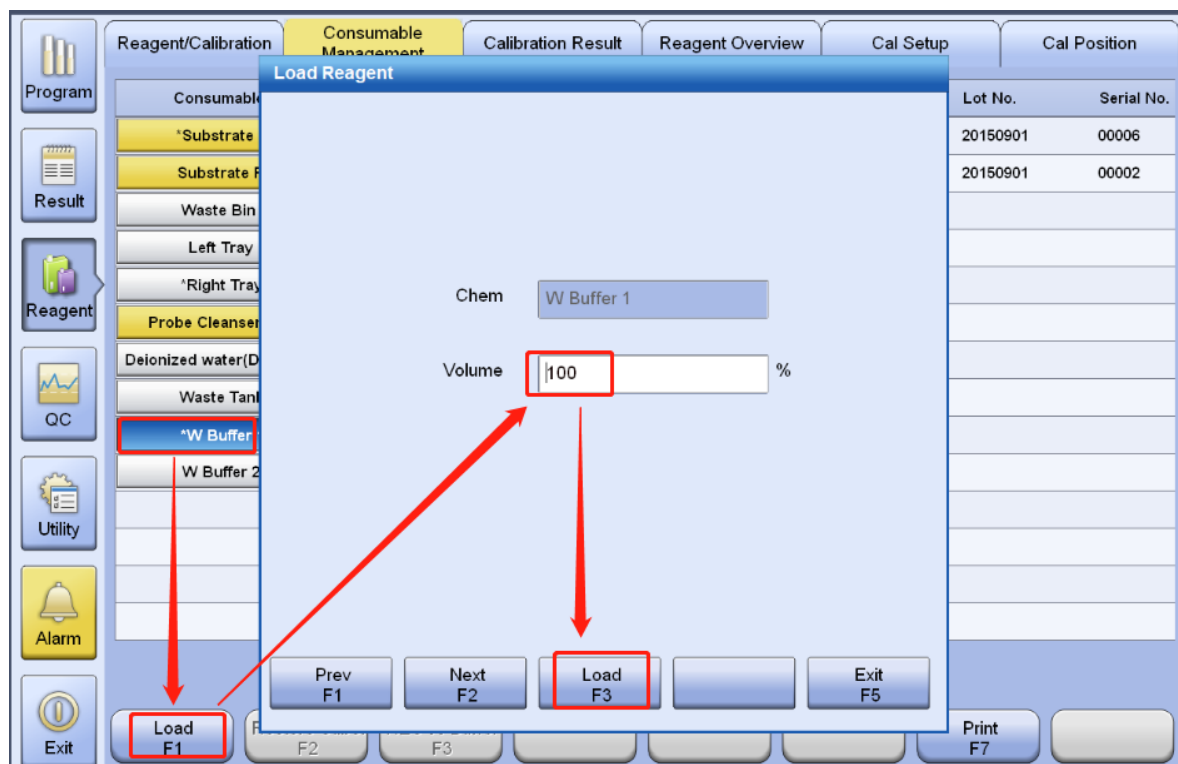


Figure3-7 Load the wash buffer

- 2) Select **Utility>Maintenance>Alignment>Fluidic Alignment and prime**; select **9. Prime Wash Buffer Tubes**, and prime the wash buffer according to the prompt information;

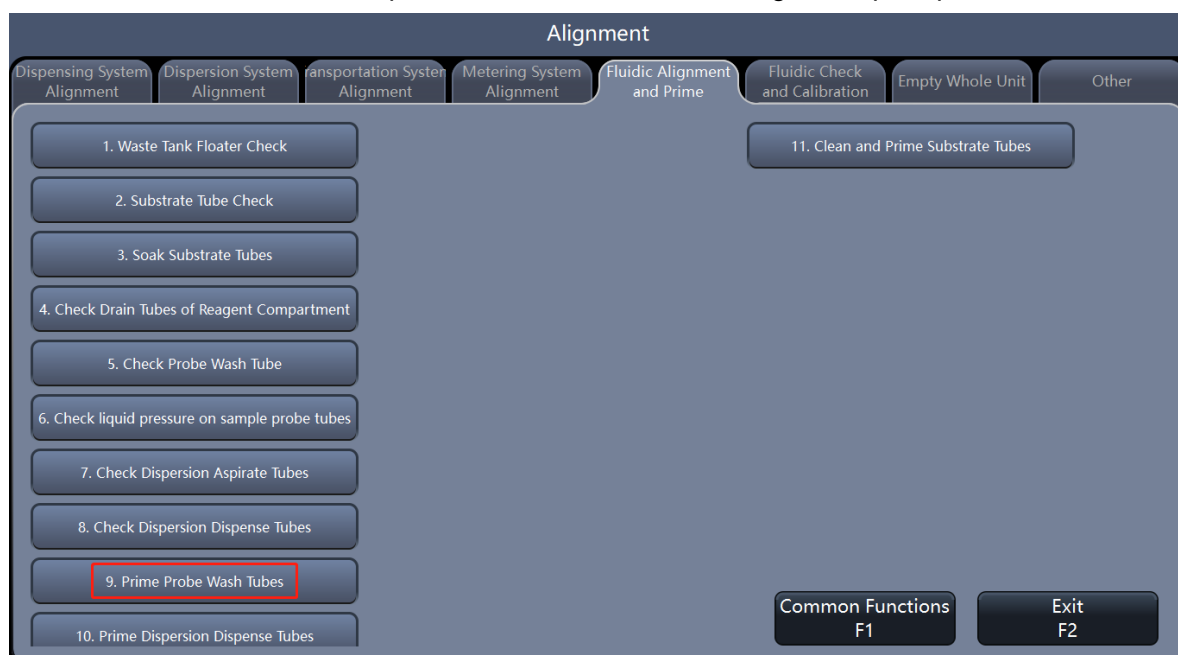


Figure 3-8 Prime wash buffer tubes

- 3) Select **Utility>Maintenance>Alignment>Fluidic Alignment and prime**;select **10. Prime Dispersion Dispense Tubes**:

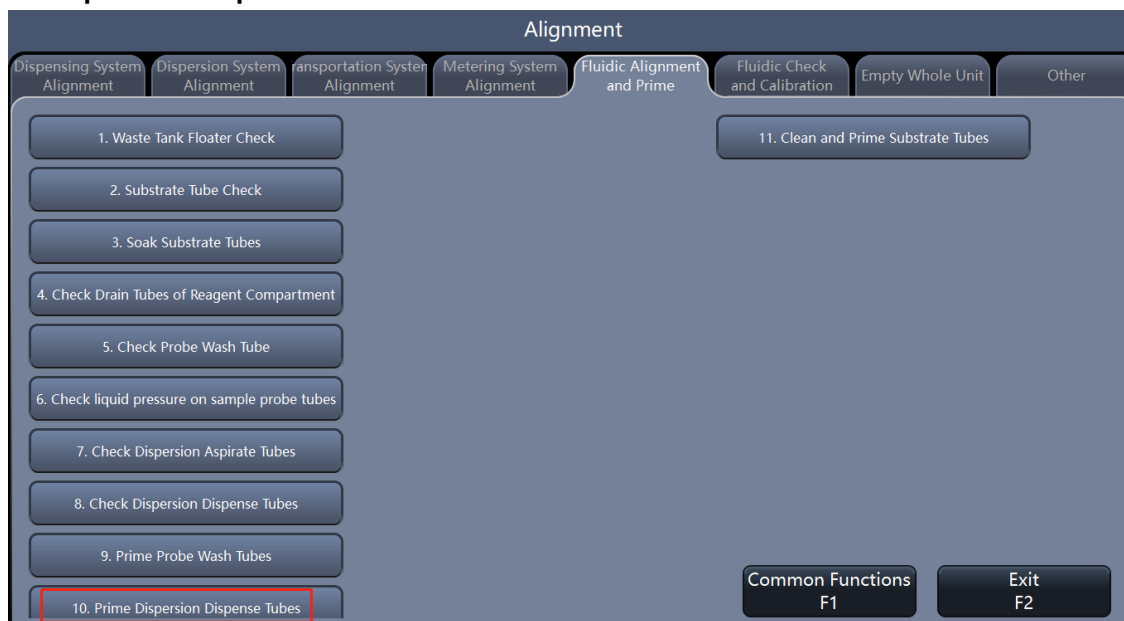


Figure 3-9 Prime Dispersion Dispense Tubes

Click "Continue", follow the "Tip/Warning" prompt , Remove the dispersion dispense probes(D1-D4) and put them in a clean container, and click "OK"

(Pay attention to the markers D1, D2, D3, and D4, and do not confuse dispense probes with aspirate probes; otherwise, overflow may be caused.)

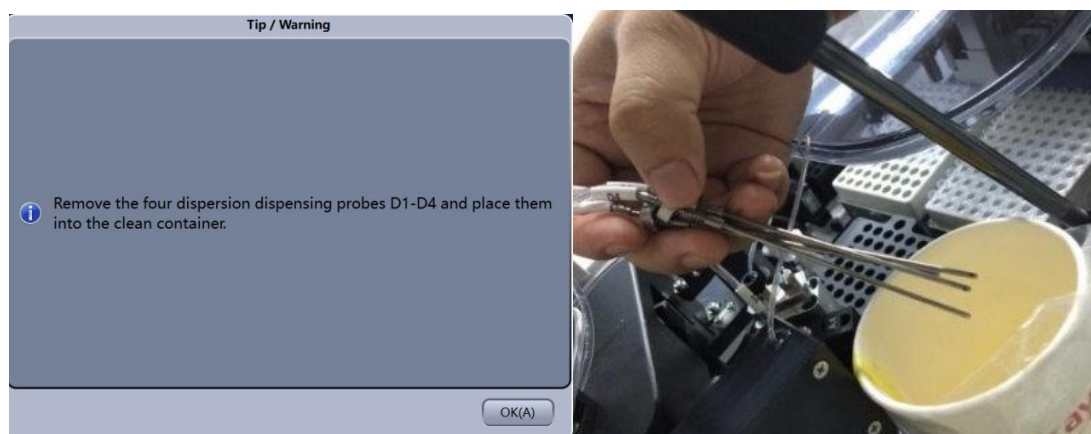


Figure 3-10 Tip Remove the D1-D4 and Place them into clean container

Click "Prime or drain (S)" to the next step, set the Execution Time to 50,Select Start to prime the dispense probes for 50 times, observe the 4-phase dispense probes till they continuously dispense wash solution without air bubbles.

Note: The prime times should not be less than 50 times otherwise the following performance may Fail

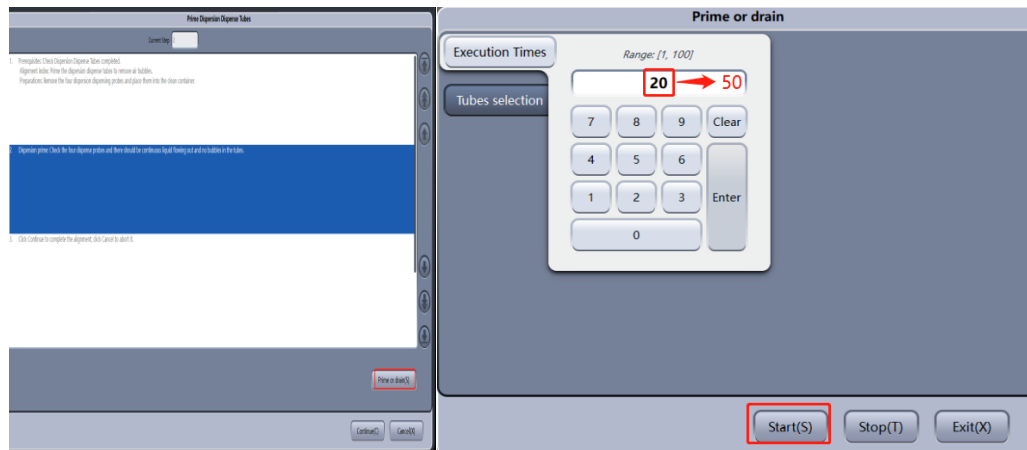


Figure 3-11 set the Execution Time to 50 Start prime

Wipe the dispense probes with clean fiber cloth and install them properly.

3.4 Original Parameter Backup

Select **Utility>Maintenance>Alignment>others>Common Function**, click **Backup and Restore of Parameters**, enters the screen for the parameter,

No	Unit	Parameter	Parameter Range	Compensation Value
1	Reagent Carousel Unit	Position Offset of Ra from Reagent Pack Home Position	(-640, 640)	0
2	Reagent Carousel Unit	Position Offset of Rb from Reagent Pack Home Position	(-224, 416)	0
3	Reagent Carousel Unit	Position Offset of Rc from Reagent Pack Home Position	(-336, 304)	0
4	Reagent Carousel Unit	Position Offset of Rd from Reagent Pack Home Position	(-400, 240)	0
5	Sample Carousel Unit	Position Offset of Counter-clockwise Movement	(-190, 180)	0
6	Sample Carousel Unit	Position Offset of Clockwise Movement	(-190, 180)	0
7	Sample Carousel Unit	Position Offset of Sample Disk 45#.	/	0
8	Sample Carousel Unit	Position Offset of Sample Disk 41#.	/	0
9	Sample Carousel Unit	Position Offset of Sample Disk 35#.	/	0
10	Sample Carousel Unit	Position Offset of Sample Disk 31#.	/	0
11	Sample Carousel Unit	Position Offset of Sample Disk 25#.	/	0
12	Sample Carousel Unit	Position Offset of Sample Disk 21#.	/	0
13	Sample Carousel Unit	Position Offset of Sample Disk 15#.	/	0
14	Sample Carousel Unit	Position Offset of Sample Disk 11#.	/	0
15	Sample Carousel Unit	Position Offset of Sample Disk 5#.	/	0

Figure 3-9 Backup and Restore of Parameters

Select **backup**, save the parameters. The parameter is located in the file:

D:\Mindray\CL900i\OperationSoft\AlignmentTool\Parameterlist. Find the parameter and name it with "date+original parameter".

NOTE: When completed alignment, must to backup the parameter again.

3.5 Main Unit Position Confirmation and Alignment

1) Preparation for position confirmation and alignment

Execute **Home**. If it enters the "Idle" state smoothly and there is no fault alarm, you can refer to the Debugging Guide to confirm and debug the key positions. Otherwise, the fault should be checked.

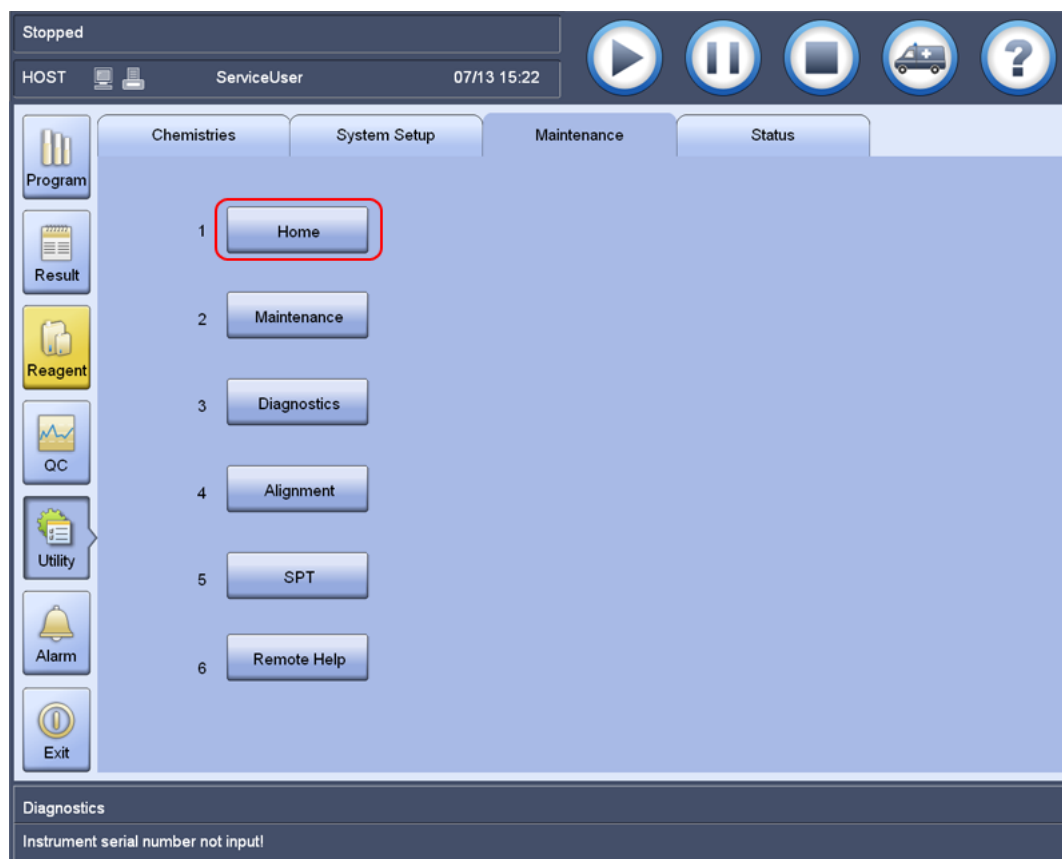


Figure3-10 Home system

2) Procedure of position confirmation and alignment

Position confirmation and alignment of probe

Select **Utility>Maintenance>Alignment>Dispensing System Alignment**

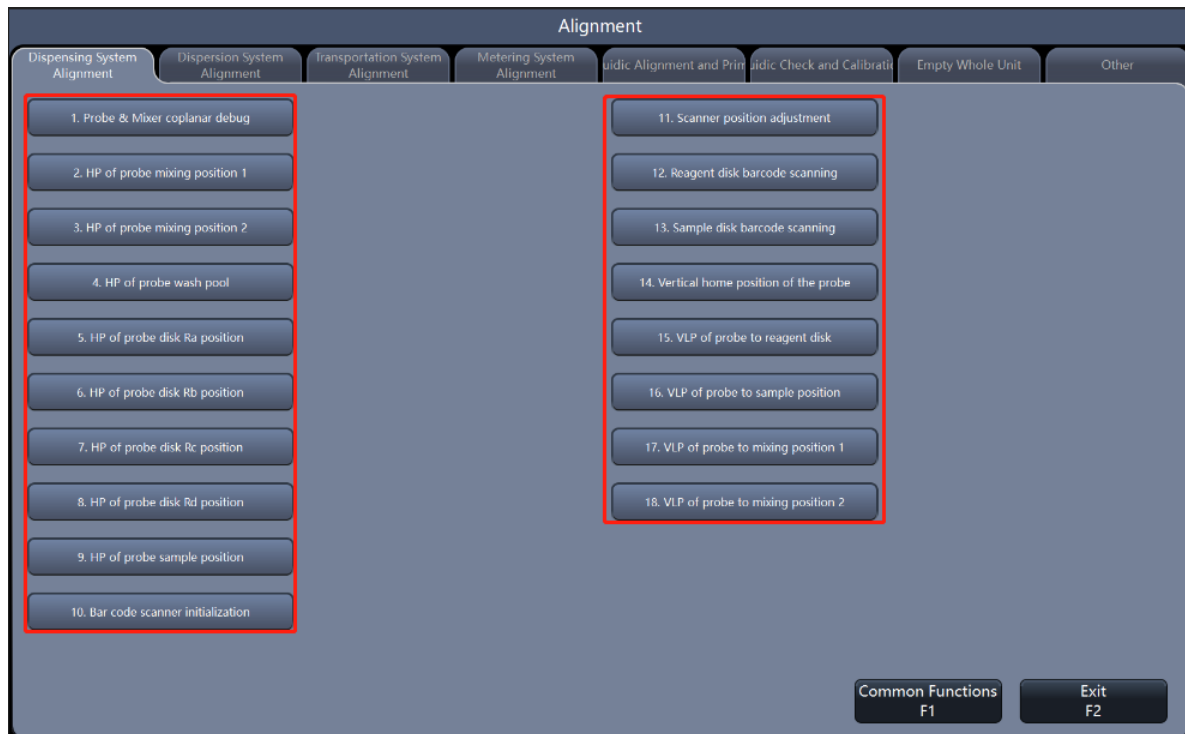


Figure3-11 Position confirmation and alignment of probe

Confirm and align the positions in the red frame above. For detailed steps, please refer to the Alignment Guide.

3) Position confirmation and alignment of gripper

Select **Utility>Maintenance>Alignment>Transport System Alignment**.



Figure3-62 Position confirmation and alignment of gripper

Confirm and align the positions in the red frame above. For detailed steps, please refer to the Alignment Guide.

4) Position alignment of dispersion system

Select **Utility>Maintenance>Alignment>Dispersion System Alignment**.



Figure3-13 Position confirmation and alignment of dispersion system

Confirm and align the positions in the red frame above. For detailed steps, please refer to the Alignment Guide.

5) Position confirmation and alignment of shielding cover

Select **Utility>Maintenance>Alignment>Photometer System Alignment**.

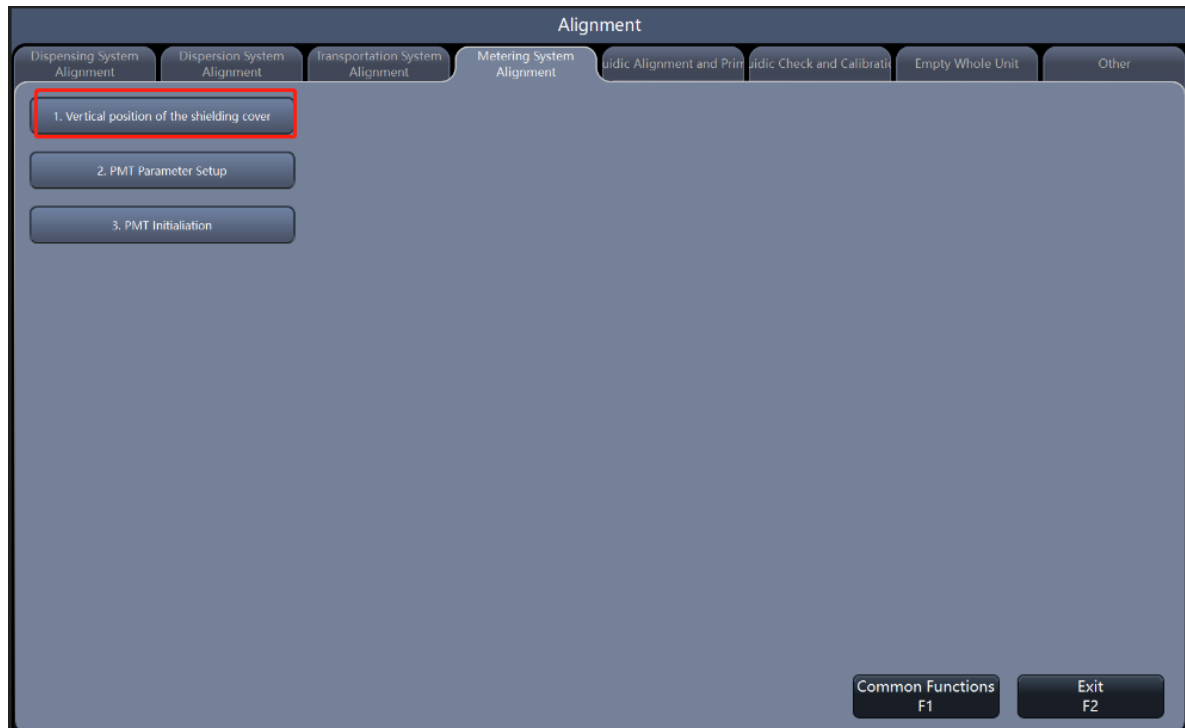


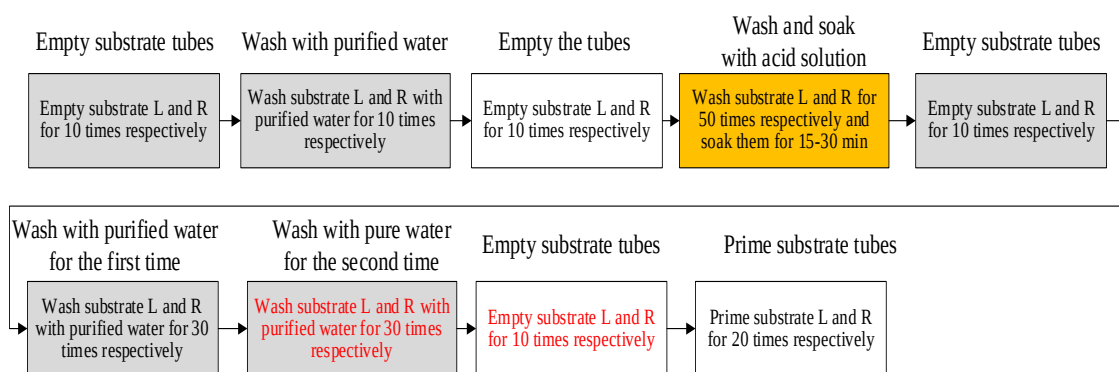
Figure3-14 Position confirmation and alignment of shielding cover

Confirm and align the positions in the red frame above. For detailed steps, please refer to the Alignment Guide.

3.6 Clean and prime substrate tubes

3.6.1 Clean substrate tube

Alignment methods and procedure:



- 1) Prepare 2 bottles of Wash solution that has be Diluted and 2 clean substrate bottles;

Note: the Wash solution is packaged in a substrate bottle, with 20ml of the acid lotion as the original solution, which needs to be diluted manually with ultra-pure water, and then diluted until the bottle is full of the substrate.

- 2) The positions of two substrate spikes are vacant;
- 3) Select Alignment -> 11. Clean and Prime Substrate Tubes to enter the alignment process..

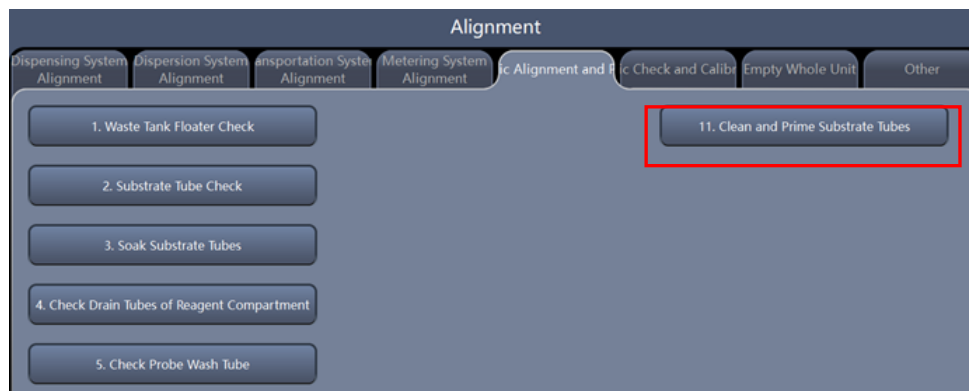


Figure3-15 Clean and Prime Substrate

- 4) Remove the substrate spike and suspend it in midair in a clean container.

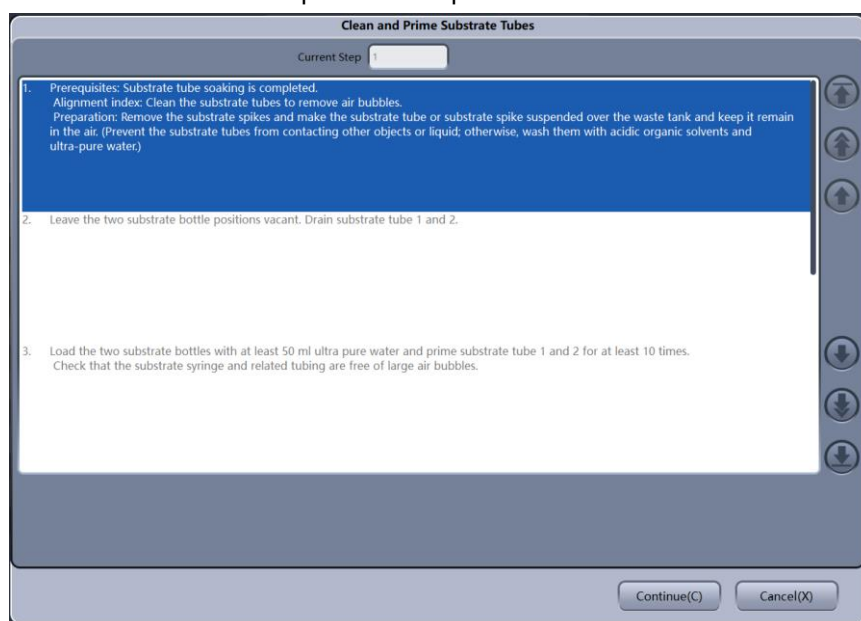


Figure3-16 suspend substrate spike in midair in a clean container

Note:

- Do not forget to remove the substrate spike. Otherwise, the liquid will be primed to the dispersion carousel, leading to abnormal results.
- Protect the substrate spikes from contamination during priming and washing.
- Remove the substrate bottles at substrate L and R, and keep the two positions vacant. Click **Continue** to proceed to the next step.

5) Empty substrate tubes

In the second step displayed on the screen, click **Prime or Drain**. The following dialog box pops up. Set the drain cycle as 10. Drain substrate L and R tubes.

Note: When the instrument is installed at the customer end for the first time, the substrate tubes are generally empty. Ignore this step and click **Continue** to go to the next step.

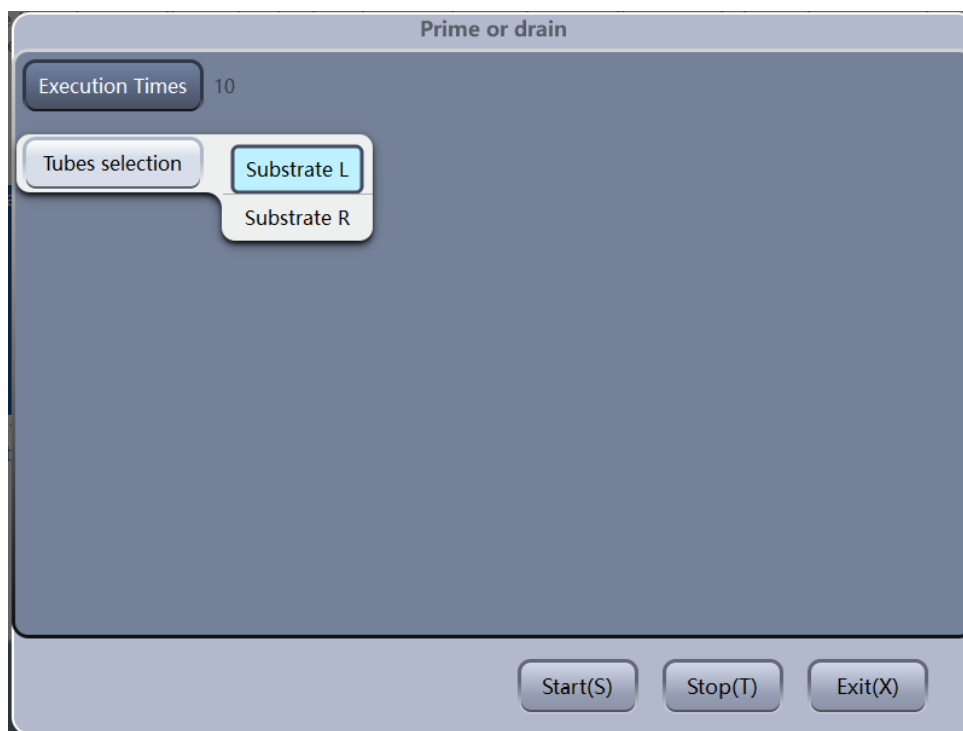


Figure3-17 Prime or drain

6) Clean substrate tubes with purified water

After drainage, click **Continue** to go to the next step of cleaning substrate tubes. Fill the clean substrate bottle with ultra-pure water and load it into the substrate L. Click **Prime or Drain** on the screen, select substrate L tube, and set the times as 10 to clean and prime the substrate L tube. Use this method to clean and prime substrate R tube.

7) Empty substrate tubes

Leave positions of substrate bottles L and R vacant, select the corresponding tubes, set the times as 10 to complete emptying the two substrate tubes.

8) Clean substrate tubes with acid wash solution

Load the acid wash solution in substrate L and R position, set the cleaning times as 50. Prime the substrate L and R respectively and **soak the substrate for 15-30 min.**

Note: The wash solution is packaged in a substrate bottle, with 20ml of the acid lotion as the original solution, which needs to be diluted manually with ultra-pure water, and then diluted until the bottle is full of the substrate.

9) Empty substrate tubes

Perform the operation still on the screen. Leave the substrate L and R positions vacant, select the corresponding tube, and set the times as 10 to drain the two substrate tubes.

10) Clean substrate tubes with ultra-pure water

The operation is still performed on the same screen. Prime each tube for 60 times (priming sequence: Substrate L tube for 30 times - > substrate R tube for 30 times - > substrate L tube for 30 times - > substrate R tube 30 times). During priming, place two bottles of purified water in the substrate container, loosen the cap, and check if the liquid column in the transparent substrate tube can aspirate liquid normally after the liquid is discharged.

11) Empty substrate tubes

Perform the operation on the same screen. Leave the substrate L and R positions vacant, select the corresponding tube, and set the times as 10 to drain the two substrate tubes.

12) Prime substrate tubes

The operation is still performed on this screen. Load an equilibrated substrate to substrate L after removing its aluminum foil at the bottom and then select **Substrate L** and input 20 times. Click **Start** to start substrate prime. After priming, use the same method to prime the tube in substrate position R.

13) Click **Continue** to go to step 4. Remove the substrate tube from the waste bottle and use dry dust-free cloth to wipe the tube. Install the substrate spike into the substrate base using one knurled small-head screw. **Avoid contacting the substrate position.**

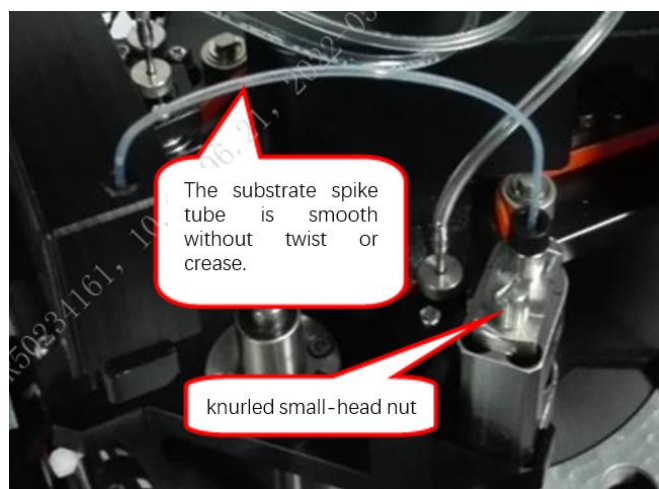


Figure3-18 Install the Substrate Spike

14) Click Continue to return to the alignment screen.

3.7 Setting up

3.7.1 Load and check the consumables

3.7.1.1 Loading the cuvette tray:

Select Left Tray, tap Load, and then pull out the left drawer assembly manually, load the cuvette tray, and push it forward after confirming that it is placed well. Load the cuvettes in the right tray with the same method.

Reagent/Calibration		Consumable Management		Calibration Result		Reagent Overview		Cal Setup		Cal Position	
Program	Consumables	Volume	Load Date	Days Left	Exp Date	Lot No.	Serial No.				
	*Substrate L	299	2017/11/11	<-99d	2015/12/31	20150901	00006				
	Substrate R	0	2017/09/29	<-99d	2015/12/31	20150901	00002				
	Waste Bin	185									
Result	Left Tray	13									
	*Right Tray	88									
	Probe Cleanser(DC)	100%	2022/04/13	<-99d	2018/11/11						
	Delonized water(DI Water)										
QC	Waste Tank	Draining									
	*W Buffer 1	1%	2017/11/14								
	W Buffer 2	99%	2017/11/14								
Reagent											
Utility											
Alarm											

Program

Result

Reagent

QC

Utility

Alarm

Exit

Load F1

Restore Subs. F2

RES W Buffer F3

Print F7

Figure3-19 Loading the cuvette tray

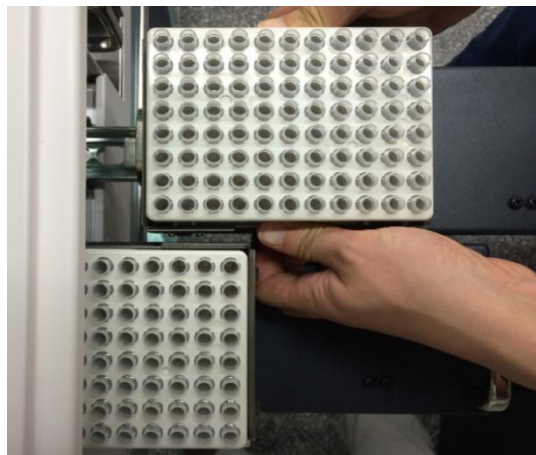


Figure3-20 Pushing it forward after confirming that the cuvettes are placed well

3.7.1.2 Load the Probe Cleanser(DC) Deionized water(DI Water) and wash buffer

Before priming the tubes, you have loaded the Probe Cleanser(DC)、Deionized water(DI Water)

and wash buffer.

3.7.1.3 Load the waste tank

Before connecting the hydropneumatic system tubes during installation, you have loaded the water tank.

3.7.1.4 Load the substrate

Before priming the substrate, you have loaded the substrate.

3.7.1.5 Besides the **Consumable Management** screen, you can query the detailed information of consumables on the **Reagent Overview** screen

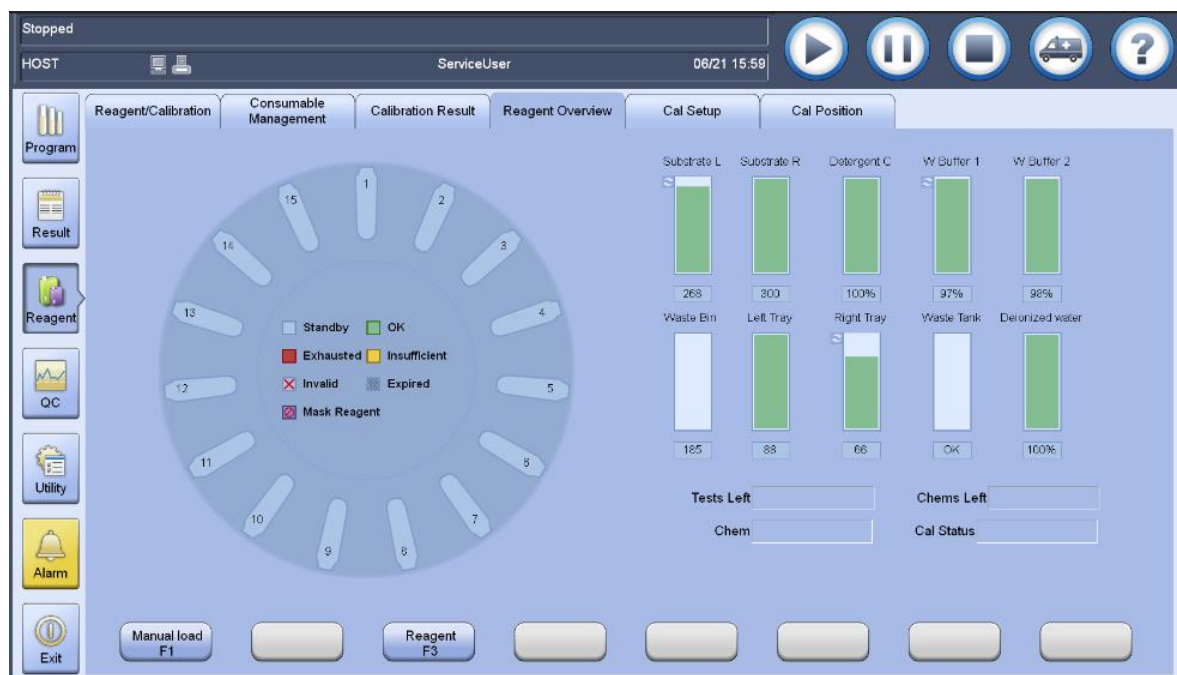


Figure3-21 Reagent overview

3.7.2 Importing and Configuring Chemistry Parameters

Select **Utility** -> **Chemistries** -> **Import**.

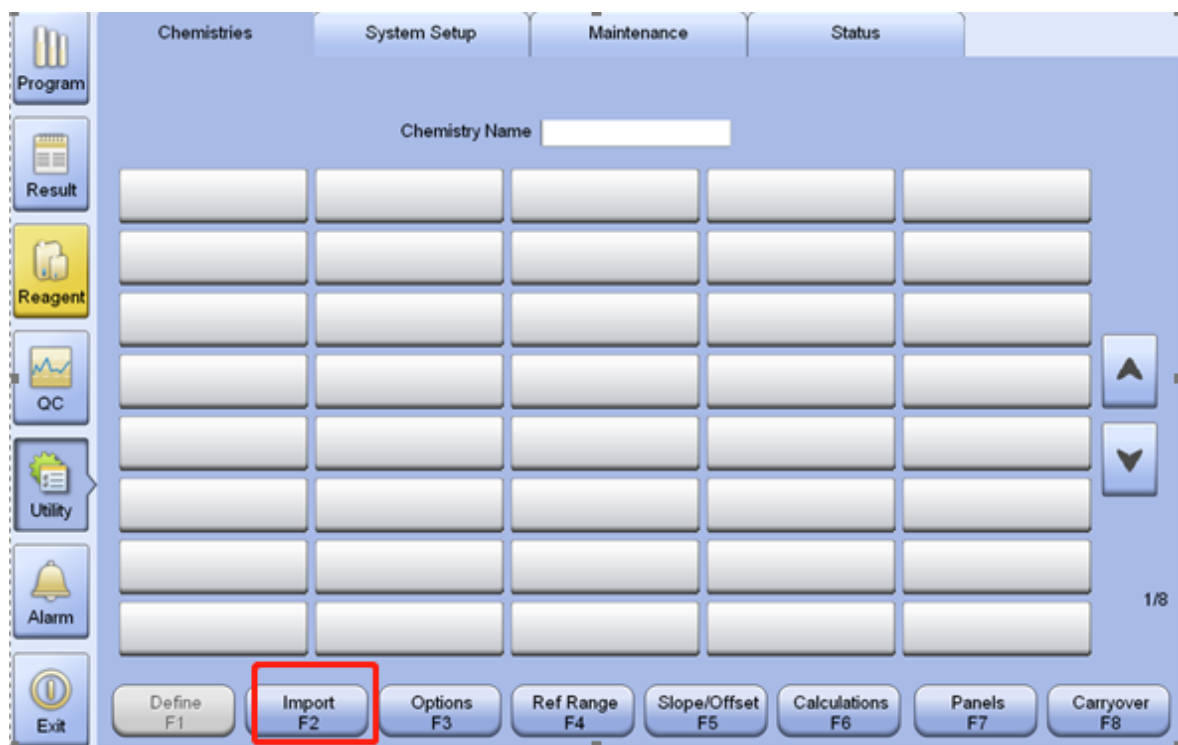


Figure3-22 Import

Select Load Default to display all Mindray reagent chemistries in the left column. Select Add All to add all chemistries to the right column.

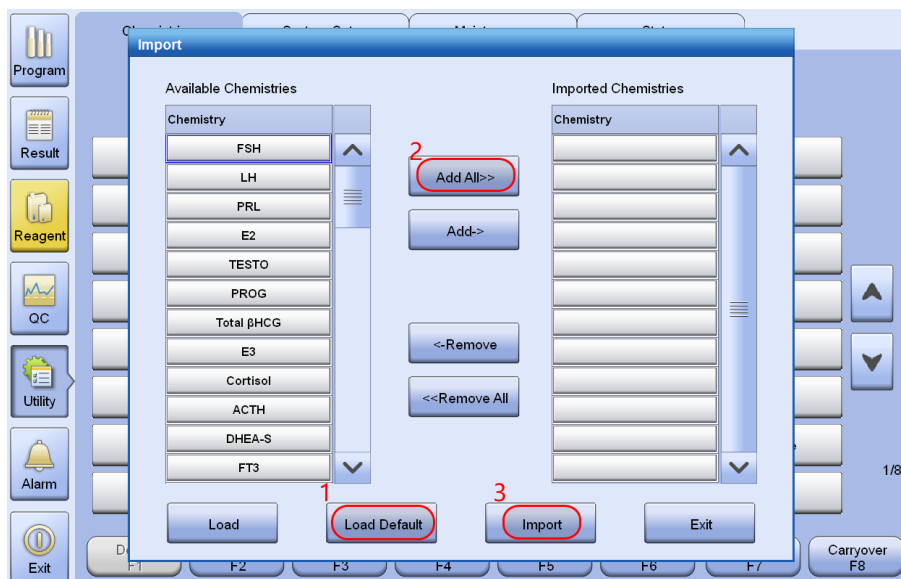


Figure3-23 Import default chemistry parameters

Select Import and then select Exit. The default chemistry parameters are imported.

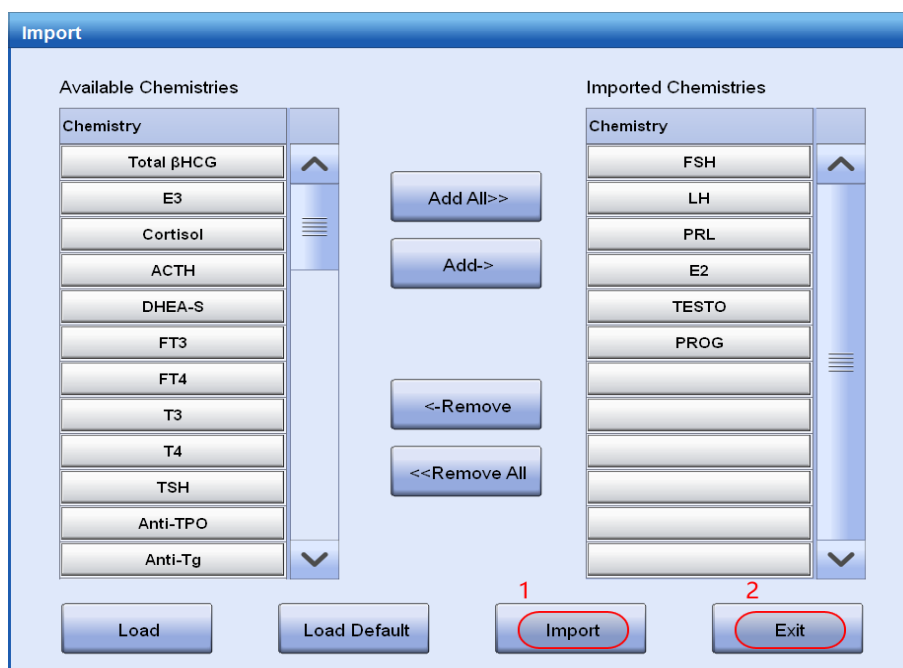


Figure3-24 Select Import and Exit

Check the imported chemistries on the Chemistries screen.

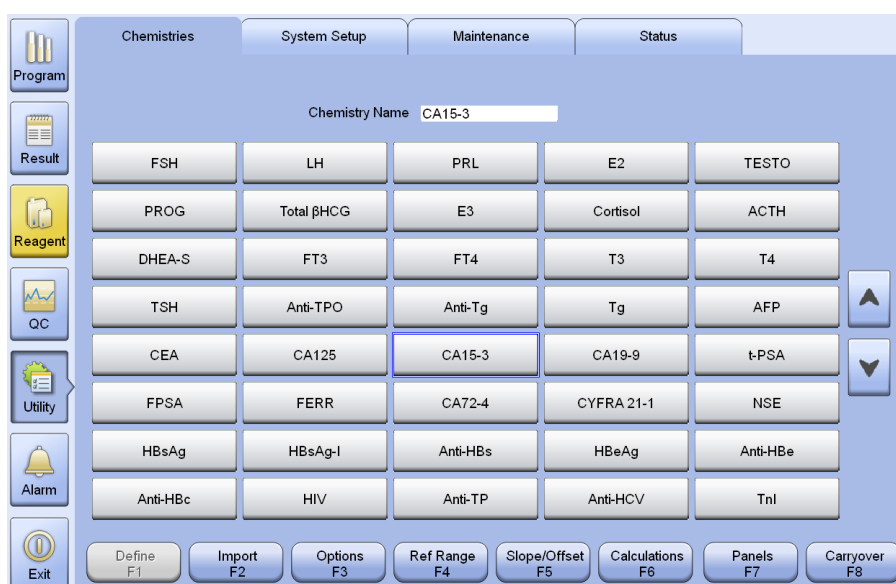


Figure3-25 Check the imported chemistries

Chapter 4 System Performance Test

4.1 DCF Diagnosis

- 1) Enter Utility→Maintenance→Diagnostics→PhotometerDiagnosis→DCF diagnosis.
- 2) Parameter settings: Set the Test Cycle as 20 and select Start to start the test.
- 3) Record the **Relative Extreme Difference**, which should be less than 1.5%.

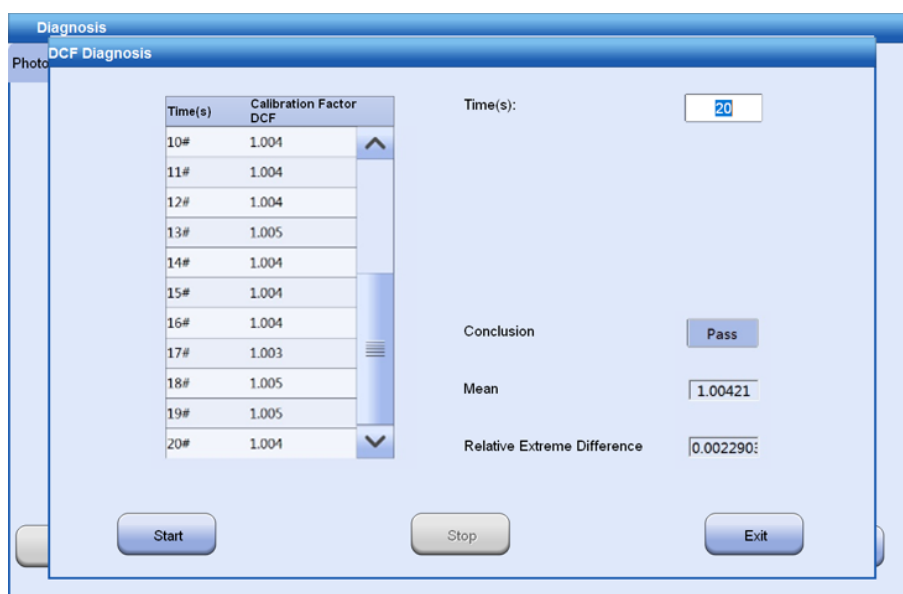


Figure4-1DCF stability test

4.2 Substrate Background detection

- 1) Select **Utility→Maintenance→ SPT→Substrate**;
- 2) Select **Substrate Background**;
- 3) Tick the **Dispersion** checkbox, tap **Start**, and wait for the results;
- 4) The instrument automatically starts detection, and displays the results after completion.

Item	index
Substrate background RLU	3000-6000
Substrate background SD	SD<100

4.3 System Repeatability

Prepare the sample: Add 1 ml System Detection Solution A to the microtube and load it to **1# sample position**;

- 1) Select **Utility > Maintenance >SPT> Photon Counting**;
- 2) Select System Repeatability;
- 3) Confirmation:

The **Sample Replicate(s)** is 20, and the **Reagent Replicate(s)** is 0;

The sample aspirate **volume (ul)** is **15**, and the sample dispense **volume (ul)** is **10**;

- 4) Tap Start, and wait for the results.
- 5) The instrument automatically starts detection, and displays the results after completion.

Item	index
Sample RLU	400000-2000000
Sample CV	$CV \leq 1.5\%$

4.4 Repeatability Test

Pick at least 1 item that will be test by user in the list below, finish the calibration at first.

Anti-Tg	Anti-TPO	LH	HCG
HBsAg	HIV	Anti-TP	Anti-HCV

Take a quality control as sample to perform 15 times repeatability test, the repeatability test should satisfy the following indices:

$$CV \leq 5\%$$

Fill the test results into the Basic Performance Test Record file attached, it will show if the repeatability test is pass or failed:



Basic

Performance Test

4.5 Original Parameter Backup

Note: After parameter backup, copy the parameter backup folder to a USB flash drive. The folder is named as (instrument SN_BackupPara) for search. It is recommended to save the parameters to the working computer or hard disk of the branch company.

Select Utility > Maintenance > Alignment > Others > Common Function, click Backup and Restore of Parameters, enters the screen for the parameter. Select Backup, save the parameters. The parameter is located in the file:

D:\Mindray\CL900i\OperationSoft\AlignmentTool\Parameterlist.

Chapter 5 LIS and Remote Help

5.1 LIS Connection

The service engineer should assist the hospital in connecting the LIS software.

Select **Utility>setting>LIS Setting**

Host Communication Parameters

Transport: Serial

IP Address: . . . Port: 7118

Serial Port: COM1

Data Bits: 8 Stop Bits: 1

Parity: None Baud Rate: 9600

Protocol: HL7 Mode: Unidirectional

☐ Auto Connect to LIS Timeout: 30 S

☐ Retry after Disconnect Interval: 30 S

☐ Send Complete Samples

☐ Send Incomplete Samples

Chem	Routine Channel No.	Dilution Channel No.
FSH		
LH		
PRL		
E2		
TESTO		
PROG		
Total β HCG		
E3		
Cortisol		
ACTH		
DHEA-S		
FT3		
FT4		
T3		

Buttons: Restore Defaults, Advanced, Connect, Save, Close

Appendix A Comparison of altitude and atmospheric pressure

Altitude	Atmospheric Pressure(kPa)	Atmospheric Pressure(psi)
-400	105.9	15.4
-300	104.8	15.2
-200	103.6	15.0
-100	102.5	14.9
0	101.3	14.7
100	100.2	14.5
200	99.1	14.4
300	98.0	14.2
400	96.9	14.0
500	95.8	13.9
600	94.7	13.7
700	93.6	13.6
800	92.6	13.4
900	91.5	13.3
1000	90.5	13.1
1100	89.4	13.0
1200	88.4	12.8
1300	87.4	12.7
1400	86.4	12.5
1500	85.4	12.4
1600	84.4	12.2
1700	83.4	12.1
1800	82.4	12.0
1900	81.5	11.8
2000	80.5	11.7
2100	79.6	11.5
2200	78.6	11.4
2300	77.7	11.3
2400	76.8	11.1

Altitude	Atmospheric Pressure(kPa)	Atmospheric Pressure(psi)
2500	75.9	11.0
2600	75.0	10.9
2700	74.1	10.7
2800	73.2	10.6
2900	72.3	10.5
3000	71.4	10.4
3100	70.6	10.2
3200	69.7	10.1
3300	68.9	10.0
3400	68.0	9.9
3500	67.2	9.7
3600	66.4	9.6
3700	65.6	9.5
3800	64.8	9.4
3900	64.0	9.3
4000	63.2	9.2
4100	62.4	9.0
4200	61.6	8.9
4300	60.8	8.8
4400	60.1	8.7
4500	59.3	8.6
4600	58.6	8.5
4700	57.8	8.4
4800	57.1	8.3
4900	56.4	8.2
5000	55.6	8.1